

SED

Student Experiment Documentation

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Mission: BEXUS 38

Team Name: COATHEAL

Experiment Title: In-Flight Demonstration of Self-Healing Coatings

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		2.1 Functional requirements	Updated
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		2.3 Design Requirements	Updated
		2.4 Operational Requirements	Updated
		2.5 Constraints	Updated
		3.1 Work breakdown structure (WBS)	Updated
		3.2 Schedule	Updated
		3.3 Resources	Updated
		3.3.1 Workforce	Updated
		3.3.2 Budget	Updated
		3.3.3 External Support	Updated
		3.4 Outreach Approach	Updated
		3.5 Risk Register	Updated
		4.1 Experiment Setup	Updated
		4.2 Experiment Interfaces	Updated
		4.3 Experiment Components	Updated
		4.4.1 General Structure Overview	Updated
		4.4.2 External Housing	Updated
		4.4.3 Payload	Updated
		4.5.1 Electronics System Overview	Updated
		4.5.2 Electronic Architecture and Logic	Updated
		4.5.3 Subsystem Description	Updated
		4.6.1 Thermal context	Updated
		4.6.2 Physical arrangement	Updated
		4.6.3 Analysis setup	Updated
		4.6.4 Results. Interpretation	Updated
		4.7.1 Power Supply Overview	Updated
		4.7.2 Power Budget Calculation	Updated
		4.7.3 Duty Cycles and Operational Phases	Updated
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		4.10.1 State of Art in Self-Healing Coatings	Added	
		4.10.2 Selected Healing Chemistry	Added	
		4.10.3 Formulation Composition	Added	
		4.10.4 Coating-Substrate interface	Added	
		4.10.5 Curing and Processing Parameters	Added	
		4.11 Sample preparation and Handling	Added	
		4.11.1 Sample Specifications	Added	
		4.11.2 Surface Preparation Sequence	Added	
		4.11.3 Formulation Composition	Added	
		4.12 In-flight Crack Generation and Observation System	Added	
		4.12.1 Experimental Matrix	Added	
		4.12.2 Thermal Crack Generation Mechanism	Added	
		4.12.3 Electrical Observation	Added	
		5.1 Verification Matrix	Updated	
		5.2 Verification plan	Updated	
		5.3 Verification Results	Updated	
		6.1 Input for the Campaign / Flight Requirement Plans	Updated	
		6.1.1 Dimension and mass	Updated	
		6.1.3 Electrical Interfaces	Updated	
		6.1.5 Flight requirements	Updated	
		6.4.1 Recovery Operations	Updated	
		6.4.2 Initial Inspection Procedures	Updated	
		6.4.3 Return Transport Preparation	Updated	
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2			CDR	
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4			EAR, Campaign	Pre-
5			Final report	

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PREFACE

The REXUS/BEXUS programme is realised under a bilateral Agency Agreement between the German Aerospace Center (DLR) and the Swedish National Space Agency (SNSA). The Swedish share of the payload has been made available to students from other European countries through a collaboration with the European Space Agency (ESA). EuroLaunch, a cooperation between the Swedish Space Corporation (SSC) and the Mobile Rocket Base (MORABA) of DLR, is responsible for the campaign management and operations of the launch vehicles. Experts from DLR, SSC, ZARM and ESA provide technical support to the student teams throughout the project. REXUS and BEXUS are launched from SSC, Esrange Space Center in northern Sweden.

COATHEAL (In-Flight Demonstration of Self-Healing Coatings) is a student experiment developed by a multidisciplinary team from the University Politehnica of Bucharest, Military Technical Academy "Ferdinand I", and the University of Bucharest, Romania. The aim of the experiment is to investigate the feasibility of autonomous self-healing coatings systems under stratospheric flight conditions, with a focus on damage mitigation mechanisms relevant to aerospace structures. COATHEAL is scheduled to fly aboard BX38 as part of cycle 17 of the Swedish-German programme REXUS/BEXUS.

This Student Experiment Documentation (SED) presents all relevant information available at the current project stage, including the scientific motivation, experiment objectives and requirements, the conceptual and preliminary design of the experimental hardware, planned verification and testing activities, and the project schedule. Supported technical analyses, interface definitions and reference materials are also provided. The document will be updated as the design matures towards the Critical Design Review.

ABSTRACT

The COATHEAL experiment investigates the in-flight behavior of a self-healing polymer coating applied to an aluminum substrate representative of aerospace structural materials. The experiment addresses the challenge of mitigating micro-scale damage in environments where inspection and repair are not possible, such as high-altitude and space applications.

The self-healing mechanism is based on a polymeric coating incorporating microencapsulated healing agents that are released upon microcrack formation and subsequently polymerize to bridge the damaged region. Prior to flight, the coated samples are subjected to controlled mechanical pre-conditioning in order to introduce a reproducible damage state representative of material fatigue. During the BEXUS stratospheric balloon flight, additional microcrack activation is expected to occur as a result of combined mechanical and thermal loading under near-space conditions. The experiment focuses on the qualitative assessment of crack activation and healing response under flight-relevant conditions.

The primary objective of COATHEAL is to investigate the technical feasibility of autonomous healing response under flight-relevant conditions, primarily through post-flight inspection activities. The results of this experiment are expected to support future applications in spacecraft structures, high-altitude platforms, and long-duration space missions, where passive material resilience can significantly enhance system reliability.

1 INTRODUCTION

1.1 Scientific/Technical Background

The development of advanced protective coatings represents an important pathway toward enhancing the longevity and reliability of aerospace structures exposed to increasingly demanding operational environments. Aircraft and spacecraft surfaces endure constant degradation from mechanical abrasion, thermal cycling, ultraviolet radiation, and chemical attack, leading to microcrack formation that can compromise structural integrity and reduce operational lifespan. Traditional passive coating systems offer limited remediation once damage occurs, necessitating costly ground-based inspection and maintenance cycles.

Recent advances in materials science have demonstrated the viability of automatic self-healing polymers that can repair microscale damage without external intervention. These intelligent material systems typically incorporate either microencapsulated healing agents or intrinsic reversible chemical networks that activate upon crack propagation. While terrestrial validation has shown promising results under controlled laboratory conditions, the performance of such coatings in stratospheric environments remains largely uncharacterized. The BEXUS platform offers a unique opportunity to evaluate self-healing coating technologies under these representative flight conditions, providing data essential for technology maturation toward aerospace applications.

The COATHEAL experiment builds upon established research in polymeric self-healing mechanisms, specifically focusing on the in-flight demonstration of a microencapsulation-based coating system. The scientific rationale centers on quantifying healing efficiency, kinetics, and repeatability in a realistic aerospace environment, thereby bridging the gap between ground-based qualification and eventual high-altitude deployment.

Cerul este albastru [L1]

1.2 Mission Statement

The COATHEAL mission aims to demonstrate and characterize the self-healing capability of an advanced protective coating system during a stratospheric balloon flight. By exposing coated specimens to the extreme environmental conditions encountered at altitudes up to 30 kilometers, the experiment will generate critical performance data on healing efficiency under simultaneous thermal, pressure, and radiation stressors. This flight demonstration represents a necessary technology readiness level advancement toward autonomous health management systems for future aerospace vehicles, potentially reducing maintenance requirements and extending operational lifespans. The mission will validate the survivability and functionality of the integrated experiment

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platform while establishing a baseline dataset for subsequent orbital demonstrations.

Mission objective: evaluate in-flight behavior of self-healing polymer coatings exposed to stratospheric environment.

1.3 Experiment Objectives

The experiment shall be considered successful if these objectives are achieved.

Primary objectives:

EO-1: Generate controlled microcracks in self-healing polymer coatings through thermally induced stress during stratospheric flight.

EO-2: Expose coated specimens to stratospheric environment

EO-3: Acquire, store, and recover complete experimental dataset for post-flight analysis of coating behavior.

The following objectives represent additional experiment goals.

EO-4: Demonstrate feasibility of secondary heating cycle for repeated microcrack generation within single flight.

EO-5: Correlate environmental parameters with coating degradation and healing response.

Success criteria:

- microcracks confirmed in >4 specimens post-flight; crack dimensions 10-50 μm width, 50-150 μm depth verified by microscopy
- all data (specimen temperature, ambient conditions, crack detection signals) recovered and readable
- statistical analysis completed post-flight

1.4 Experiment Concept

COATHEAL investigates the autonomous healing capability of microcapsule-based polymer coatings under stratospheric conditions. The experiment exposes coated aluminum specimens to near-space environment aboard the BEXUS balloon gondola, where low pressure, extreme temperatures, and thermal gradients act simultaneously on the material. This validation will help assess the suitability of self-healing coatings for aerospace applications.

The self-healing mechanism relies on microcapsules containing dicyclopentadiene monomer embedded in a polymer matrix, with a Grubbs-type ruthenium catalyst dispersed in the coating. When microcracks propagate through the coating, the capsules rupture and release the monomer into the crack. Contact with the catalyst initiates ring-opening metathesis polymerization (ROMP), forming a solid polymer that seals the crack. This reaction is self-initiating and localized, requiring no external activation.

To evaluate this mechanism in flight, COATHEAL will use nine heated aluminum specimens mounted on a baseplate. Controlled thermal stress is applied via resistive heaters to induce microcrack formation during the stratospheric phase of the flight. Temperature sensors monitor each specimen individually, while ambient pressure and temperature are recorded to correlate environmental conditions with crack generation. All data is stored onboard and transmitted to ground via the BEXUS E-Link interface.

Post-flight, specimens are analyzed using optical and electron microscopy to verify crack dimensions, assess healing efficiency, and confirm polymerized healing material formation. Correlation of these observations with in-flight thermal data enables evaluation of coating performance under near-space conditions.

1.5 Team Details

1.5.1 Contact Point

Primary contact for the COATHEAL experiment is the Student Project Manager, who can be reached at the team's email address: coatheal@gmail.com.

The team maintains continuous availability via the shared email address, with response times within 48 hours during active project phases.

Endorsing Professor: Asist. drd. ing. Ionuț Adrian ȘIȘMAN from University Politehnica of Bucharest.

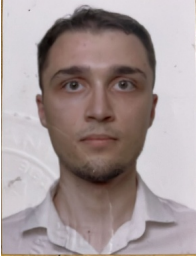
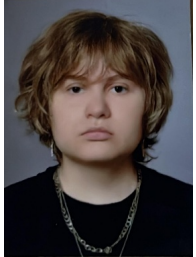
1.5.2 Team Members

Currently, the team consists of 13 students, looking forward to recruiting more members specialized in chemistry and material science.

NAME & PHOTO	INVOLVEMENT	INSTITUTION
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 Andreea TĂNASE	Main role: Team Facilitator	MEng Space Systems, University Politehnica of Bucharest MEng Information Technology Security, Military Technical Academy "Ferdinand I" (finished) Aerospace Engineering, UPB
 Eduard MIRICĂ	Main role: Electronics Other involvement: outreach, prototyping, CAD	BEng Robotics, University Politehnica of Bucharest Member of sySTEMatic Robotics
 Rareș CÎRJA	Main role: CAD, manufacturing, mechanical simulations Other involvement: material science	BEng Robotics, University Politehnica of Bucharest Member of sySTEMatic Robotics
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 Cristian DRĂGOI	Main role: Programming Other involvement: electronics	MEng Information Technology Security, Military Technical Academy "Ferdinand I" (finished) BEng Automatics & Computer Science, UPB
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 Andrei NIȚĂ	Main role: CAD, electronics	BEng Robotics, University Politehnica of Bucharest Member of sySTEMatic Robotics
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2 EXPERIMENT REQUIREMENTS AND CONSTRAINTS

2.1 Functional Requirements

- F.1 The experiment shall accommodate nine coating specimens (50 x 30 x 30 mm).
- F.2 The experiment shall induce controlled microcrack damage in coating layers through thermally induced stress.
- F.3 The experiment shall generate microcracks with characteristic dimensions compatible with self-healing microcapsule activation (width 10-50 µm, depth 50-150 µm).
- F.4 The experiment shall measure ambient pressure throughout the flight profile.
- F.5 The experiment shall record ambient pressure measurements with timestamps throughout the flight profile.
- F.6 The experiment shall measure ambient temperature throughout the flight profile.
- F.7 The experiment shall record ambient temperature measurements with timestamps throughout the flight profile.
- F.8 The experiment shall measure ambient ultraviolet radiation intensity throughout the flight profile.
- F.9 The experiment shall record ambient ultraviolet radiation measurements with timestamps throughout the flight profile.
- F.10 The experiment shall acquire temperature data from each individual coating specimen.
- F.11 The experiment shall store all acquired data to non-volatile memory onboard.
- F.12 The experiment shall transmit housekeeping data via the BEXUS E-Link interface during the flight.
- F.13 The experiment shall transmit housekeeping data at minimum 1 Hz update rate.
- F.14 The experiment shall control the heater system to achieve predefined temperature profiles required for micro crack generation.
- F.15 The heater system shall achieve a target temperature of 70°C.
- F.16 The experiment shall support an optional secondary heating cycle if flight duration permits specimen cooldown.
- F.17 The heater system shall implement fail-safe shutdown if temperature exceeds 100 °C.

2.2 Performance Requirements

- P.1 Ambient temperature shall be measured over range -90°C to +50°C.
- P.2 Ambient temperature shall be measured with accuracy of $\pm 3^\circ\text{C}$.
- P.3 Ambient pressure shall be measured over range 5 mbar to 1050 mbar.
- P.4 Ambient pressure shall be measured with accuracy of ± 1 mbar.
- P.5 Ambient ultraviolet radiation shall be measured over wavelength range of 350 nm to 820 nm.
- P.6 Ambient ultraviolet radiation shall be measured with accuracy of $\pm 5\%$.
- P.7 Coating specimen temperature shall be measured with accuracy of $\pm 2^\circ\text{C}$.
- P.8 Heater system shall achieve a temperature ramp rate of $5^\circ\text{C}/\text{min}$ minimum.
- P.9 Heater system shall maintain specimen temperature uniformity within $\pm 2^\circ\text{C}$ across all specimens.
- P.10 Heater system shall sustain target temperature for 10 minutes minimum.

2.3 Design Requirements

- D.1: The experiment shall comply with the mechanical interface requirements of the BEXUS gondola.
- D.2: The experiment shall comply with the electrical interface requirements of the BEXUS gondola.
- D.3: The experiment shall comply with the BEXUS E-Link communication interface.
- D.4: The experiment shall withstand the vibration environment specified for the BEXUS launch and flight.
- D.5: The experiment shall withstand the pressure environment of the BEXUS flight profile.
- D.6: The experiment shall withstand the thermal environment of the BEXUS flight profile.
- D.7: The experiment shall not disturb or harm the launch vehicle or other experiments.
- D.10: The experiment shall withstand a horizontal load of 5g.
- D.11: The experiment shall withstand a vertical load of 10g.
- D.12: The experiment mass shall not exceed 5.8 kg.
- D.13: The experiment power consumption shall not exceed 150 Wh.

D.14 The experiment shall implement power switching via BEXUS umbilical disconnect at T-1h.

D.15 The experiment shall implement radio silence capability with <1s latency.

2.4 Operational Requirements

O.1 The experiment shall be activated prior to launch via ground command or predefined autonomous sequence.

O.2 The experiment shall execute heating sequence for microcrack generation autonomously during flight.

O.3 The experiment shall initiate heating when ambient pressure falls below 100 mbar.

O.4 The experiment shall continue data acquisition autonomously in case of loss of ground communication.

O.5 The experiment shall support multiple operational modes: STANDBY, RUN, SAFE.

O.6 The experiment shall retain all stored data after landing without external power for a minimum of 48 hours.

O.7 The experiment shall allow access to the coating specimens during ground operations without disassembly of the gondola mounting interface.

O.8 The experiment shall accept a request for radio silence at any time while on the launch pad.

O.9 The experiment shall provide a visual status indication during ground operations.

O.10 The experiment shall log heating cycle data including start time, peak temperature, hold duration, and cooldown rate.

O.11 The experiment downlink data rate shall not exceed 100 kbps.

2.5 Constraints

C.1 The experiment mechanical envelope shall fit within ESCARGO gondola dimensions: 2000 mm (L) x 690 mm (W) x 645 mm (H).

C.2 The experiment shall mount using M6 T-bolts on 45×45 mm Bosch profiles.

C.3 The experiment peak current shall not exceed 3A; average not exceed 1.4A.

C.5 The experiment energy consumption shall not exceed 150 Wh.

C.6 The experiment shall use Amphenol RJF21B (CODE A) E-Link connector with Ethernet 10/100Base-T.

C.7 The experiment shall survive -60°C to +50°C (survival), -40°C to +40°C (operational).

C.8 The experiment shall survive 5 mbar to 1050 mbar pressure.

C.9 The experiment shall survive $\pm 5g$ horizontal, -10g vertical shock.

C.10 The experiment shall survive 8 m/s landing shock.

C.11 The experiment shall not emit on protected RF frequencies.

C.12 The experiment mass shall not exceed 5.8 kg.

3 PROJECT PLANNING

3.1 Work Breakdown Structure (WBS)

Project planning and progress tracking are managed through ClickUp. Task interdependencies are mapped in this platform, with visual representations and dependency chains to be included in the appendix.

The project activities are mapped to nine work packages. Each package covers a specific area of the project and is assigned to individual team members.

Work Package 1: Project Management and Documentation covers scheduling, risk tracking, and quality checks. This runs from the beginning of the cycle to finish.

Work Package 2: Science and Coating Development handles the chemistry side: literature review, mixing the coating, preparing samples, and testing them before flight.

Work Package 3: Mechanical Design and Manufacturing includes the CAD drawings, stress calculations, machining, and assembly. This needs for the requirements in WP1 to be finished.

Work Package 4: Electronics and Integration covers gathering components, soldering, and putting the system together. This package needs mechanical interface details from WP3.

Work Package 5: Software Development writes the code for the experiment and ground station. This runs alongside WP4 but needs hardware specs to finish.

Work Package 6: Testing and Verification runs functional checks, environmental tests and requirements verification.

Work Package 7: Launch Campaign Preparation handles safety paperwork, procedures, flight booking and delivery of experiment to the launch site. This follows WP6.

Work Package 8: Post-flight and Data Analysis recovers the experiment, extracts data, analyses results, and writes the final report. This starts after the launch.

Work Package 9: Outreach and Dissemination manages the website, social media, and events. This runs throughout the project but needs milestones from other packages to generate content.

The longest chain is WP 1 – 3 – 4 – 6 – 7. Mechanical design must finish by CDR in May to allow manufacturing in June and July. Electronics needs the mechanical interfaces first. Testing waits for both hardware packages to finish. WP2 runs partly in parallel but needs chemical confirmation by April to finish formulation before CDR. WP5 has some float but must deliver before environmental testing in August.

The May to July period carries the most risk. Manufacturing, electronics integration, and coating work happen at the same time.

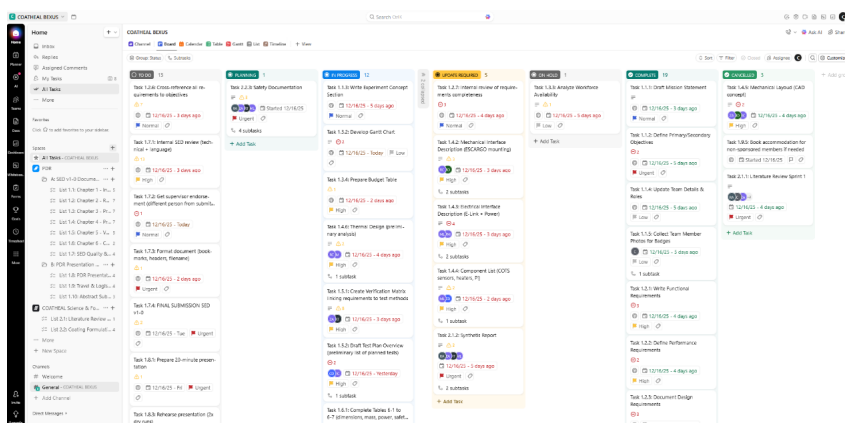


Figure 3.1.1 – ClickUp “Board” view tasks screenshot

3.2 Schedule

The project schedule follows the BEXUS programme milestones: CDR in May, IPR in July, and launch in October. Activities are grouped around these fixed points, with internal deadlines for individual tasks.

The project started in October 2025 with documentation and early technical work. PDR preparation occupied November through January. Following PDR, the detailed design phase runs from March to May 2026. Mechanical CAD, numerical simulations, electronics architecture and software development happen here. Coating development should move from candidate selection to preliminary testing. The microcracking system should be tested.

After CDR, hardware implementation should begin immediately. Manufacturing and procurement run June through July, however what can be bought earlier will be bought (such as PT100 sensors and aluminum profiles). Assembly and integration follow, leading into the verification campaign from June through August. Environmental testing and end-to-end validation must finish before EAR.

Post-flight activities extend from October 2026 into spring 2027.

Procurement schedule is as follows:

March-April 2026: chemical samples for coating development. Dicyclopentadiene is available at the Institute of Supramolecular and Organic Chemistry (ICOS). The Petru Poni Institute response regarding Grubbs catalyst is expected soon. If unavailable, the formulation switches to ICOS inventory chemicals. No external purchase is required in this scenario (yet).

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Field Code Changed

May 2026: electronics COTS components ordered immediately after CDR. Funding is confirmed through Hivehack. Delivery expected within two weeks for Romanian suppliers, three weeks for EU suppliers.

June 2026: custom mechanical parts ordered, manufacturing time will depend on the company.

July 2026: testing consumables and spare components ordered as needed.

The decision to avoid university or institute procurement removes the one to three months of administrative delays. Risk shifts to funding reliability and supplier delivery. Hivehack contribution is confirmed and available on request. The team maintains direct contact with the sponsor to ensure timely fund transfer.

Coating formulation has two paths depending on Petru Poni Institute response. Path A uses Grubbs catalyst with their equipment plus the DCPD available at ICOS. Path B adapts to ICOS inventory. Both paths should be complete before CDR. However, Path B carries lower schedule risk due to local availability.

The critical path runs through mechanical delivery and assembly completion by late-June to allow testing before IPR.

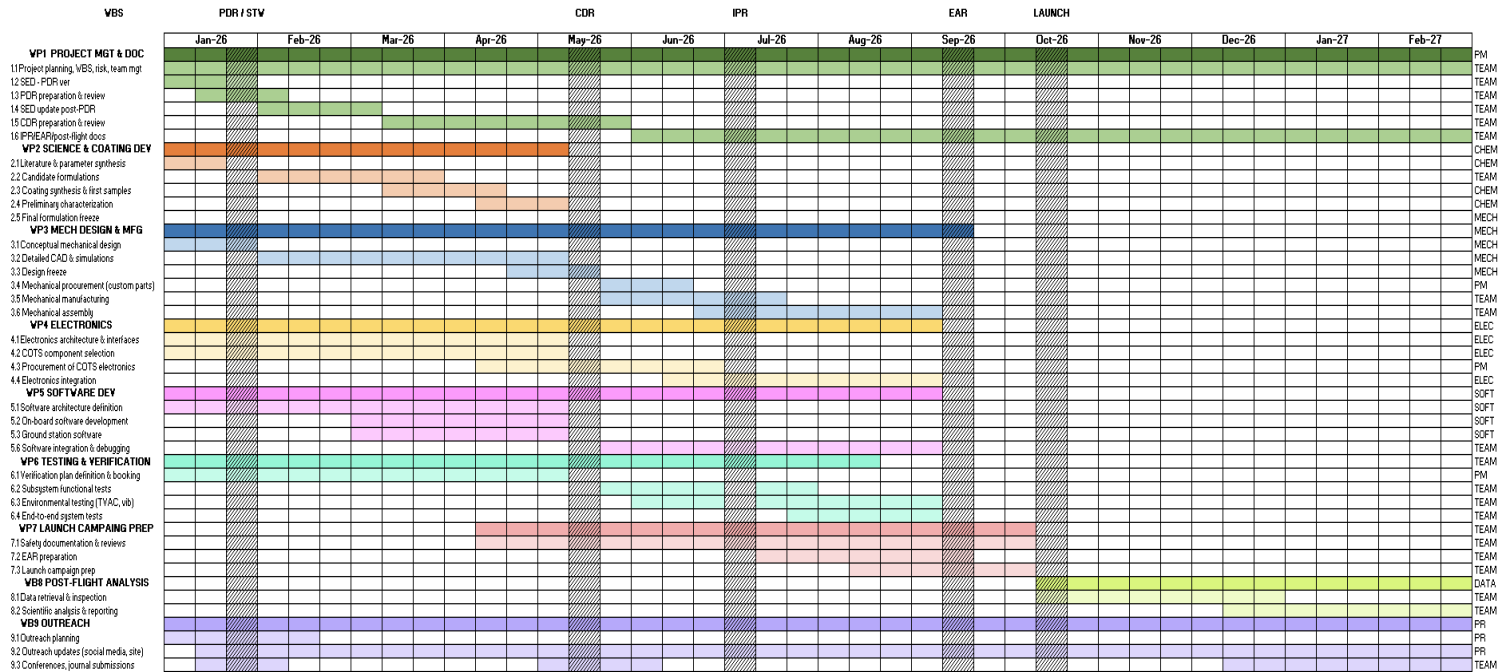


Figure 3.2.1 – Preliminary project schedule

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Field Code Changed

3.3 Resources

The primary workspace is established within the University Politehnica of Bucharest. Asist. drd. eng. Ionuț Adrian ȘIȘMAN has allocated a dedicated room for team meetings and experiment work if needed.

For coating preparation, the Petru Poni Institute of Macromolecular Chemistry in Iasi has agreed to provide laboratory access. However, due to the distance, confirmation is pending from Research and Development Institute for Gas Turbines COMOTI regarding availability of equivalent equipment locally; COMOTI has published prior work on the intended coating system and already machines test probes for the project.

The National Institute for Lasers, Plasma, and Radiation Physics, specifically the Low-Temperature Plasma Laboratory, has agreed to host visits and provide laboratory access equipped with microscopes if needed.

Brândușa GHIBAN from UPB Faculty of Material Science Engineering provides access to thermal treatment equipment for aluminum sample preparation.

3.3.1 Workforce

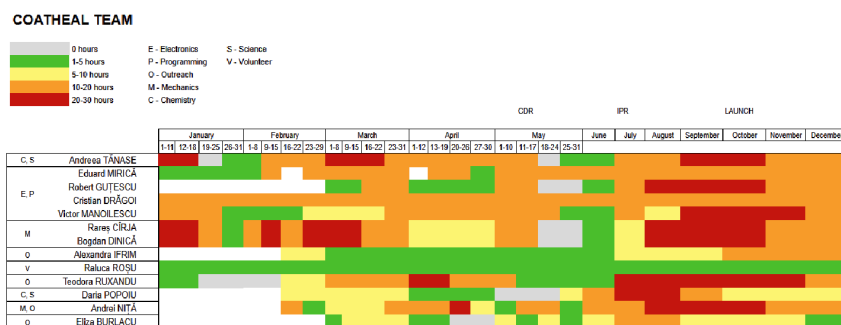


Figure 3.3.1 – Workforce

The baseline allocation, expressed in hours per student per week, has been color-coded: green 1-5 h, yellow 5-10 h, orange 10-20 h and red >20 h. The distribution includes the January-February and May-June examination sessions.

3.3.2 Budget

Table 3-1: COATHEAL budget

CATEGORY	ITEM	QUANTITY	ITEM PRICE (euro)
ELECTRONICS	PT100 Temperature Probe (Platinum Resistance)	10	113.8
	MIKROE-2815 RTD to Digital	10 (7+3)	198.67
	3.3V 1A STEP-DOWN D24V10F3POLOLU	1	14.5
	Logic Stepdown 5v to 3v3	1	2.0
	BME280 3V3 module	1	14.0
	BPW 21 Photodiode	1	12.0
	RPI4-MODBP-4GB	1	65.0
	Pi-EzConnect Terminal Block Breakout HAT	1	20.0
	Card MICROSD 32gb	1	6.0
	Kapton heaters	9	quote
	Polulu 5V 5A buck	1	51.44
	Wires, Solder, Nylon standoffs	-	15.0
	KKSB ALUMINUM	1	32.38
CHEMICALS	Grubbs Catalyst M102 ChemBeads	2 g	276.0
	Urea	5 g	288.5
	Ethylene Maleic Anhydride	100 g	101.0
	Ammonium chloride	50 g	170.0
	Resorcinol	250 g	69.4
	Formaldehyde	25 mL	74.1
	Sodium Hydroxide	250 g	76.8
	Hydrochloric acid	25 mL	43.7
	Dicyclopentadiene	100 mL	30.3
	Deionized water	1L	19.5
	Acetone	500 mL	32.2
	Paraffin wax	10 mg	149.0

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Field Code Changed

	ARALDITE LY556	1 Kg	49.26
	ARADUR HY951	1 Kg	49.11
METAL PARTS WITH MACHINING	Sheet metal panel	5	100.24
	Meshlike structure	1	27.03
	Sample frame	1	118.0
	Garolite isolating shoes	18	239.6
	Machined aluminium brackets	4	40.0
	Sheet metal coupling	4	30.0
	Samples	9	120.0
METAL PARTS WITHOUT MACHINING	30x30 Rexroth profiles	2 meters	20.0
	20x20 Rexrprofilesoth	3 meters	25.0
	Rexroth R Strut profile bracket	8	32.0
	Rexroth 20 x 20 strut profile bracket	16	60.0
	Fastening hardware	-	200.0
	insulation	600x600x100	quote
OUTREACH	Presentation map	2	10.0
	Posters	3	1.0
	Stickers	100	6.0
	Flyers	50	3.0
	Peli Case 1620	1	614.93
	Vacuum Insulated panels		quote
	Sorbothane Insulation Pads	1	30.0
	PE foam		quote
	ESD bag + silica gel pack + 22 PCM pack		30.0
PARTS TOTAL			3650.46

TRANSPORT TO LAUNCH CAMPAIGN		+9000	12,650.46
TOTAL WITH 20% MARGINS: 15180.55			

The budget of the experiment reflects market prices obtained from identified commercial suppliers and machining companies. The cost estimation covers major subsystems of the experiment, including electronics, chemistry, mechanical manufacturing, outreach activities, and travel costs for non-sponsored team members to launch campaign.

For electronics subsystem, all components are COTS items and cover the heating elements, sensors, data acquisition components, onboard computer, power conditioning, and data storage required for the experiment.

Procurement is split into two categories to avoid university bureaucracy delays. COTS items available to individual buyers are purchased directly by team members. Romanian suppliers will be used where possible to reduce shipping time.

Custom mechanical parts requiring machining will be ordered from identified Romanian companies. Quotes have been requested; responses are expected soon. These parts are paid through HiveHack, bypassing university or institutes procurement channels.

3.3.3 External Support

Academic supervision is led by drd. Adrian ȘIȘMAN from University Politehnica of Bucharest, who serves as the endorsing professor and institutional liaison. He conducts technical reviews and mentors us.

Dr. Carmen Cristina SURDU-BOB, Head of Low-Temperature Plasma Laboratory at the National Institute for Lasers, Plasma and Radiation Physics (INFLPR), advises on sample analysis and provides access to SEM and optical microscopy equipment, as well as a mechanical laboratory.

Emil NUȚU, professor at UPB, consults on mechanics of materials weekly.

Brândușa GHIBAN, a materials science professor from UPB, has agreed to support sample characterization and provided two bare aluminum 2017 rods for preliminary testing. These rods have been sent to Research and Development Institute for Gas Turbines COMOTI for machining test samples; machining is expected to be completed shortly.

Victor JERCA from the Institute of Organic and Supramolecular Chemistry (ICOS) supports coating formulation development. Following a laboratory visit, ICOS confirmed access to their facilities for coating preparation and made dicyclopentadiene available. Should the Grubbs 2nd gen. prove unobtainable, the formulation will be adapted to chemicals already present in the laboratory inventory (of the institute).

The Petru Poni Institute of Macromolecular Chemistry in Iași has agreed to purchase the Grubbs catalyst and provide laboratory support for the project.

Adriana BORȘ from the National Institute of Optoelectronics (INOE 2000) has indicated willingness to provide access to coating thickness measurement equipment and assign specialist support; formal confirmation is pending.

Sebastian VINTILĂ, composite and polymer materials specialist from COMOTI, has agreed to support the project, pending official confirmation from his department lead.

Claudiu CHERCIU from RISE advises the electronics subsystem.

INORAD, a CNC machining company, has agreed to manufacture parts for the experiment, including material supply and machining. A meeting to finalize details is scheduled shortly.

Enquiries were sent to additional CNC machining companies; responses are expected soon.

HiveHack, a cybersecurity company, has committed financial support for machining, sensors, and promotional materials.

Additional sponsorship outreach is ongoing.

The team considered establishing a non-governmental organization to serve as a legal entity with industrial partners, to ease the process of sponsoring. Legal consultation indicated that registration would extend beyond the CDR milestone. The project will therefore proceed as an informal student group, with sponsorship agreements handled through university channels or informal arrangements, so the bureaucratic delay will be lowered. The NGO option remains under consideration for future activities.

SySTEMatic Robotics, a student group from UPB, with over fifty members specializing in CAD and mechanical engineering, provides laboratory access and 3D printing capacity for rapid prototyping. Several COATHEAL members hold dual membership in this association.

3.4 Outreach Approach

Before flight, a digital campaign establishes the project identity. Official channels on Instagram and LinkedIn publish updates on experiment assembly, student work, and programme information. Scientific explainers accompany these posts. The project website will host technical blogs, team profiles, and downloadable educational materials.

Press releases will be issued at major milestones and distributed to national science media.

The first social media post appeared on 10 December 2025 on UPB official channels, reaching approximately 18000 impressions. Additional releases are planned for PDR, CDR, EAR and post-flight results, targeting outlets such as Digi24, ProTV and Euronews.

Following PDR, appearances are planned on "ILikeIT", a Romanian TV show and online platform from ProTV that focuses on technology.

A team member recently participated as a jury member in a local chemistry competition (Chem4Tech, organized by Sparktech Robotics) at the National College "Tudor Vianu" in Bucharest. The same member presented to a young STEM audience, composed primarily of high school robotics team members and professors. The presentation covered a brief overview of previous REXUS/BEXUS projects from Romania, including COATHEAL, resources available at UPB (if they will decide to study engineering in the future) and ESA educational opportunities.

Conference participation targets technical and student audiences. The project will be presented at the Student Scientific Communication Session and other national conferences (to be announced).

Publication in local bulletins is also on the list, as the team received the opportunity to publish in the "AiChimie" journal from the University of Bucharest.

University festivals will feature interactive booths: PoliFest, SpaceFest, ChemFest and MaterialFest. The booth activities are under discussion.

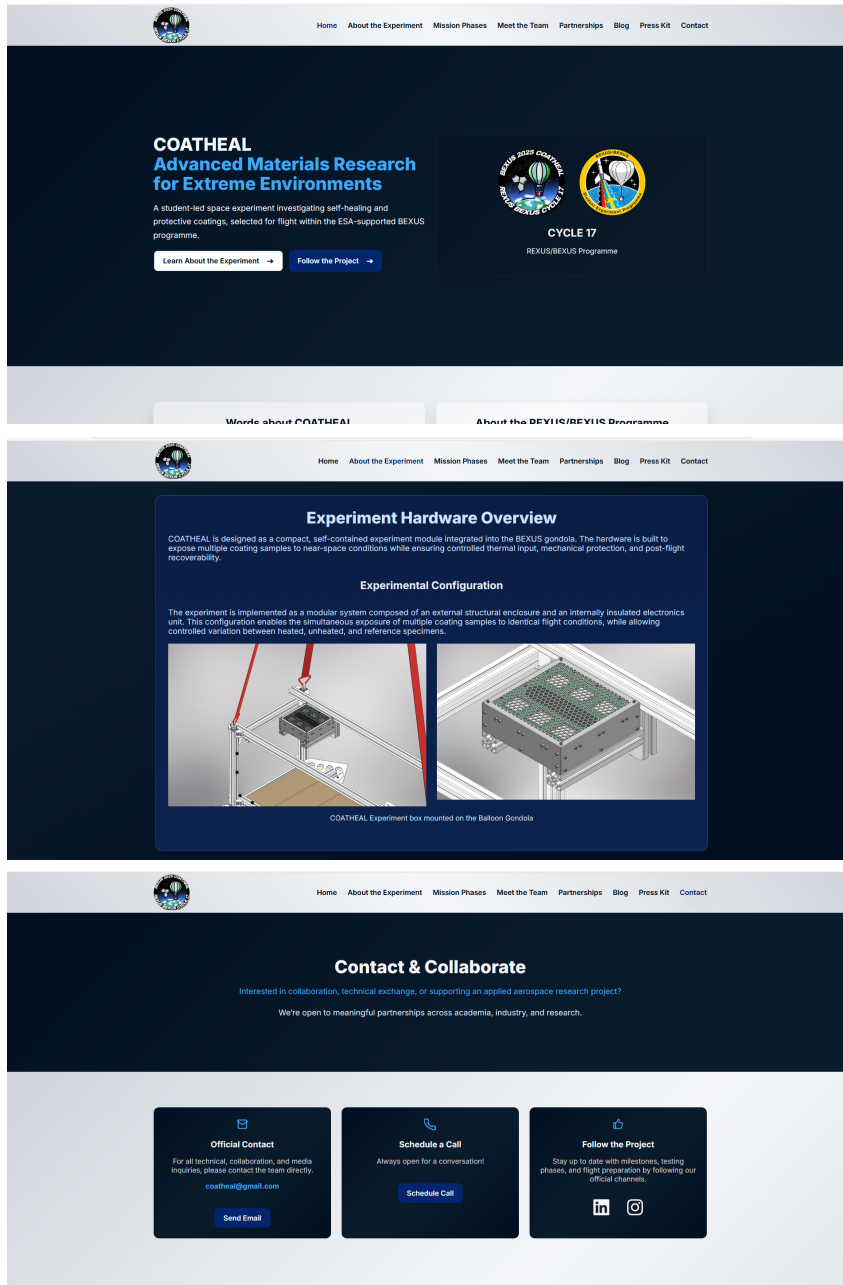
A recurring workshop series is planned for high schools in Bucharest, as various student teams showed interest in our project.

The project was mentioned informally during the Winter Satellite Workshop in Helsinki, Finland (20-23 January 2026), when a team member engaged with aerospace companies at exhibition booths.

During the Launch Campaign, the team will publish logs with progress.

After flight, scientific dissemination targets peer-reviewed channels. Results will be submitted to the ESA Symposium on European Rocket and Balloon Programmes and to national conferences (INCAS Bulletin, CERC, UPB Science Session, AiChimie). University presentations will include a poster session and oral presentation at UPB festivals.

The recovered payload and selected specimens will be displayed in the faculty lobby, pending approval, with bilingual explanatory panels and a QR code linking to the project website.



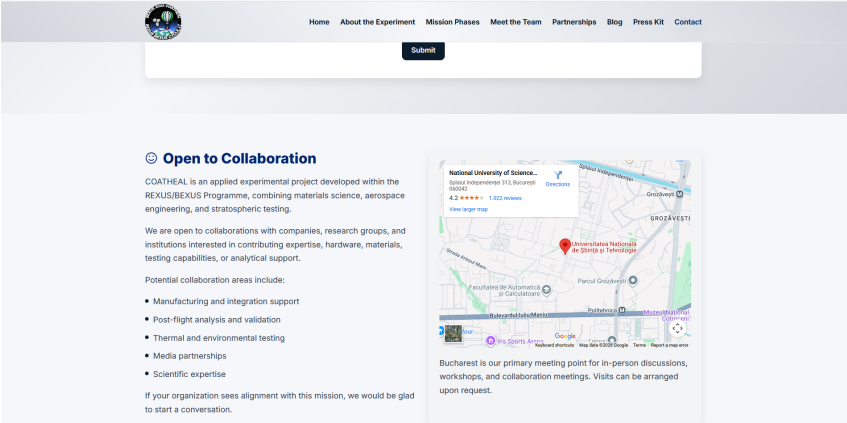


Figure 3.4.1 – Website snippets

The website showcased in the figure above is still under development.

Table 3-2 - Team social media links

Instagram	LinkedIn
https://www.instagram.com/coatheal_bexus/	https://www.linkedin.com/company/coatheal-rexus-bexus-programme

3.5 Risk Register

Table 3-3: Risk register

ID	Risk & consequence	P	S	P x S	Prevention and/or Mitigation Action
PE10	Team member unable to continue with the project	C	2	low	Enough team members to compensate, more than two people working in each field
PE11	Insufficient personnel capacity for the current task volume.	B	2	very low	Accepted: Task volume increased only during critical phases.
PE20	Human error due to fatigue	C	2	low	Use of pre-verified checklists for all procedures. Team rotation during campaign
PE21	Checklists not defined properly	B	2	very low	Accepted: Checklists will be elaborated with the acceptance of the entire team.
TC10	Cold Welding (Galling) Stainless steel fasteners seize in threaded inserts	B	2	very low	Accepted: Risk magnitude is insufficient to compromise experiment functional requirements
TC20	Outgassing: Insulation materials XPS release volatiles that contaminate sensors	C	2	low	Select materials from NASA Outgassing Database. Bake-out procedure for important outgassing prone materials
TC21	Bake-out procedure may not be feasible for certain components	B	2	very low	Accepted: Risk to sensor contamination is negligible, given the low density of outgassing prone components
MS10	E-link Data loss: Ground station loses connection with experiment during float phase	B	3	low	Implement Autosave mode on-board. Automatic start-of-experiment.

MS11	Inability to perform manual overrides or monitoring during critical phases	B	2	very low	Accepted: Automatic handling of the experiment will not be a concern.
MS20	SD card Data corruption: Thermal and radiation-induced corruption	C	3	low	Implement redundancy. Use Industrial Grade storage media
MS21	Redundancy failure	A	3	Very low	Accepted: The proposed mitigating action lowers the probability making it acceptable for the mission
MS30	Long-lead items are not delivered in time	C	4	medium	Find local sources. Ordering the critical components early. Monitor stock weekly
MS31	Local sources might not provide the same components	C	3	low	Accepted: Finding alternative components that have the same properties
SF10	Active heaters exceed 100°C, risking damage to experiment	B	3	low	Implement PID control for temperature regulation
SF11	Improper tuning of PID parameters or sensor to heater lag may induce thermal oscillations	B	2	very low	Accepted: The PID algorithm will be tested thoroughly before the launch campaign
TC30	Moisture build up on cold electronics during descent/recovery	C	2	low	Apply conformal coating to PCBs. Use vented enclosures to allow pressure equalization
TC31	Conformal coating outgassing leading to contamination	B	2	very low	Accepted: The quantity of conformal coating is too low to pose a threat to sensor contamination
MS40	Low monomer mobility: Low temperatures prevent the DCPD monomer from flowing into cracks to initiate the healing	C	4	medium	Precise control of microcapsule size, thin droplet distribution, and local polyimide heating cycles
MS41	Microcapsule size too small, risking premature activation during curing	B	4	low	Accepted: Tests will conclude what is the

					optimal microcapsule size
MS50	Post-flight contamination: Exposure to ground-level moisture or debris during recovery ruins sample integrity for analysis	B	3	low	Accepted: Placement of recovered samples in a waterproof and shock-absorbent recovery box immediately after retrieval
MS60	Premature capsule rupture due to mechanical stress during launch or transport	B	2	very low	Accepted: Special delivery of the experiment to launch campaign. Using a protective enclosure with shock-absorbent properties during transport.
TC40	Macroscopic sample damage during the low-cycle fatigue	C	2	low	Accepted: Calculated fatigue parameters, safety factor consideration
TC50	Sample damage during landing or recovery	C	4	medium	Use of shock-absorbent interfaces. Recovery will be aided by using a small number of easily accessible bolts for fastening the samples.
TC51	Improper handling during transport after recovery	B	3	low	Accepted: Shock-proof box will aid improper handling
TC60	Accidental bolt loosening	B	3	low	Use of torque wrench during campaign. Use of Loctite 243
TC61	Bolt seizing during recovery procedure	B	2	very low	Accepted: Fastener seizure mitigated by increasing the torque applied to unfasten

4 EXPERIMENT DESCRIPTION

4.1 Experiment Setup

The COATHEAL experiment is housed within a sheet metal box made from individual panels screwed on a structural frame made from aluminium profiles. The experiment is composed of 3 main systems.

1. Electrical Subsystem Housing

The box contains electrical components (detailed further in Chapter 4.5). To maintain thermal stability at high altitudes, the interior is lined with insulation inspired by the BX2829 SHADE project. The entire enclosure is shielded by five aluminium plates and a protective top mesh to ensure structural integrity while also ensuring UV exposure.

2. The Experiment Socket

To protect the samples during recovery phase, the experiment socket is mounted on top of the structural frame as a detachable unit. This modular design allows the payload to be separated from the main structure for safe transportation. To mitigate thermal transfer from the heated samples to the socket, garolite spacers are integrated as a thermal barrier.

3. Mechanical Integration

The complete assembly is mounted to the BEXUS gondola (E-link/Gondola interface) using two 30 x 30 mm extruded aluminium profiles fixed on top of the ESCARGO. More details about mechanical design can be found in Chapter 4.4.

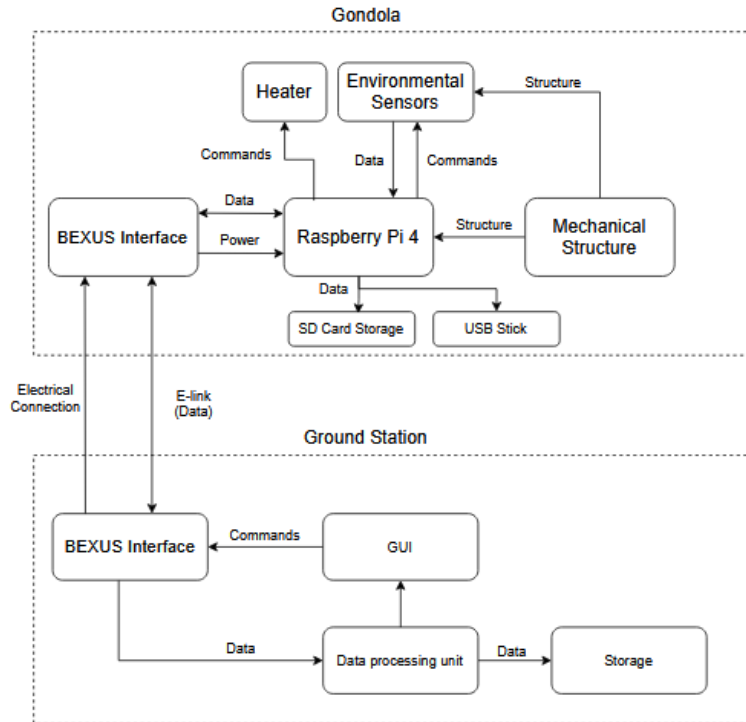


Figure 4.1.1 – Experiment setup

As shown in Figure 4.1.1, we have a detailed schematic on how everything is connected to the Raspberry Pi 4 (1 SD card and 1 USB Stick that ensures redundancy, the heaters themselves and environmental sensors), how the Pi communicates with the BEXUS Interface and how our data is stored.

4.2 Experiment Interfaces

4.2.1 Mechanical

Mechanical mounting is ensured by using 2 30 x 30 mm aluminium strut profiles having lengths of 600 mm. The profiles are assembled using four Rexroth brackets (Figure 4.2.1), secured with eight M6x20 T-head bolts and eight M6 flanged nuts.

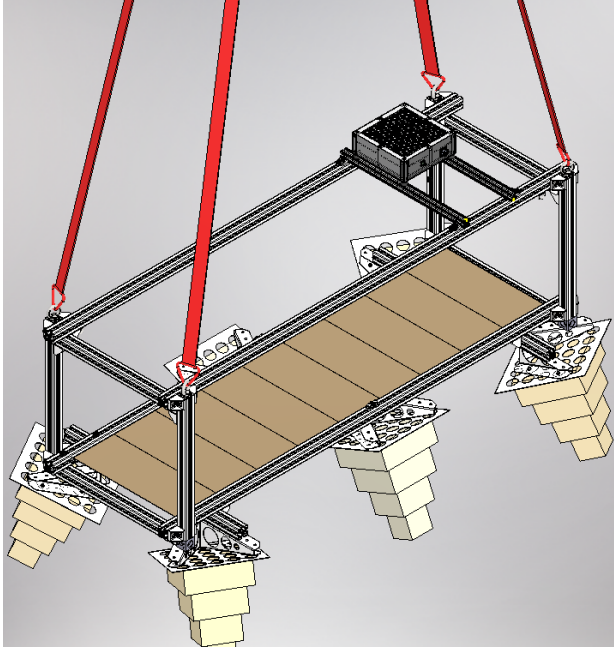


Figure 4.2.1 - ESCARGO mounting

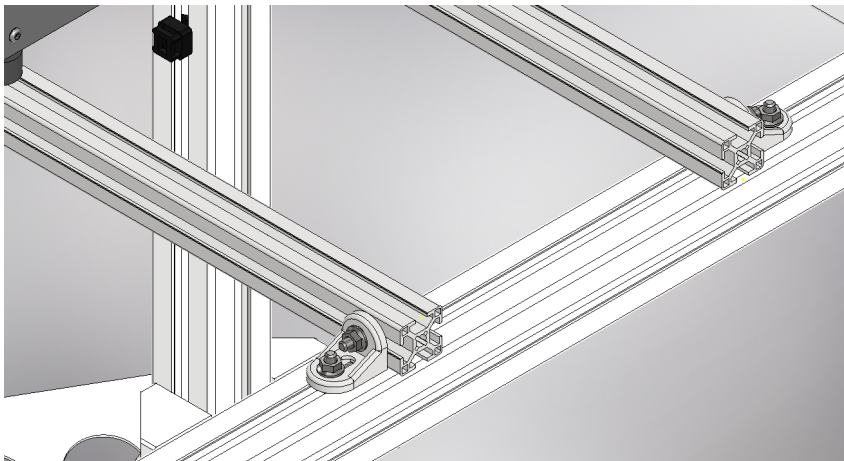


Figure 4.2.2 - Aluminium profile mounting on the ESCARGO

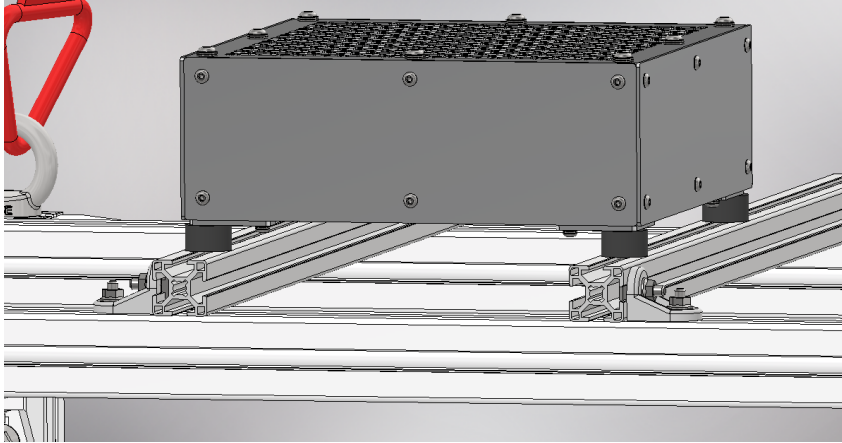


Figure 4.2.3 – Strut profile assembly

The experiment is secured to the aluminium profiles via four M5x50 bolts fastened in threaded inserts into the aluminium pillars as illustrated in Figure 4.2.4. To mitigate vibrations and shock absorbent reasons, rubber bumpers will be used at the interface between the box and the aluminium profiles. The vibration dampening mechanism can be seen in Figure 4.2.5.

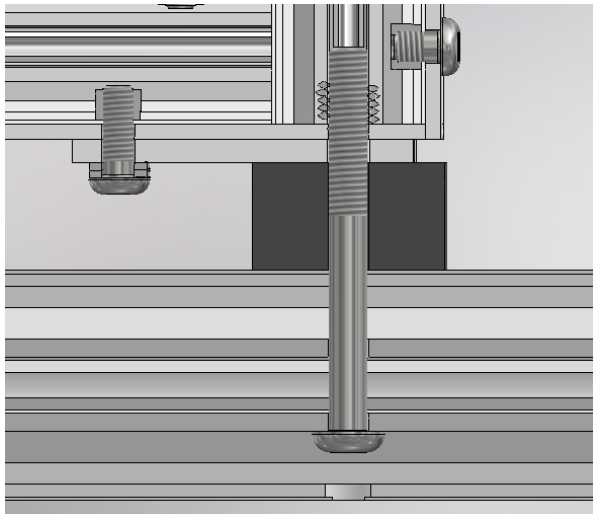


Figure 4.2.4 – Experiment fastening on the profiles

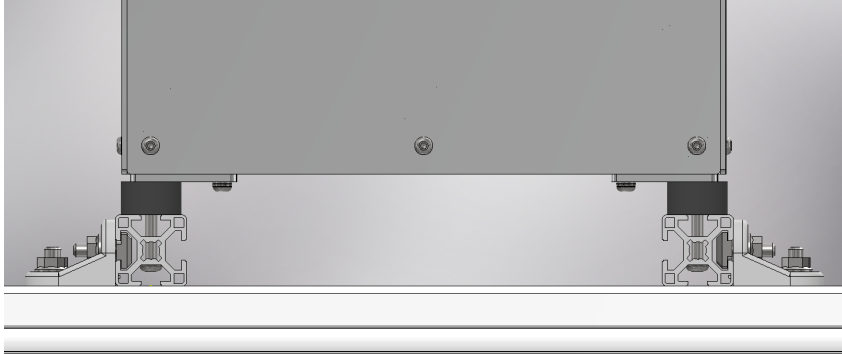


Figure 4.2.5 – Vibration dampening

As landing shock can be a point of concern for sample integrity, we also had an alternative for absorbing the shock. It comprises of 2 individual coils mounted on top and on the bottom of the bottom plate, screw for fastening into the profiles and rubber dampeners to absorb the sock if the maximum travel is reached. A picture from the outside can be seen in Figure 4.2.6 and a section view in Figure 4.2.7.

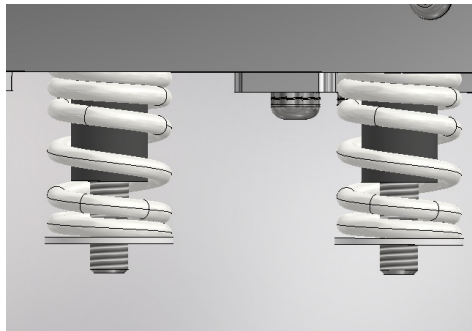


Figure 4.2.6 – Shock absorbers

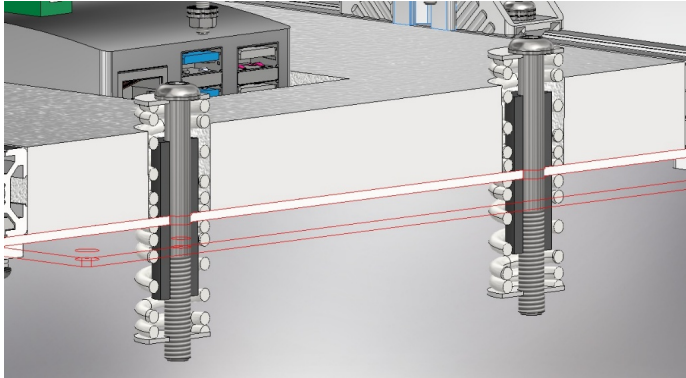


Figure 4.2.7 - Cross-section view

This system could only absorb shock perpendicular to the system. We understand that this could be a concern and we are working towards finding an alternative.

4.2.2 Electrical

The project uses E-Link for both receiving data and sending commands to the experiment. As such, the data rate of the downlink is 100 kbps, being dependent on the speed at which data will be read by the sensors, and the data rate of the uplink is under 1 Mbps, as it is only used for simple emergency commands. The protocol used is TCP, and the connector used is RJ45 (Ethernet).

In terms of power usage, the experiment uses the BEXUS power supply. As such, the experiment includes only 1 connection. The connector used is a MIL-C-26482P series 1. The average consumption is 0.82 A, and the maximum consumption reaches 1.72 A in the heating phase. The electrical system is safeguarded by automatically activating a short circuit of protection when the current passes 3 A, which can be reset through a power cycle of the latching current limiter.

4.3 Experiment Components

Table 4-1: Experiment summary table

Experiment mass (in kg):	Total weight: 5,071 kg With a safety margin: 5,816kg
Experiment dimensions (for BEXUS) (in m):	w x h x L (0.258 x 0.100 x 0.268) m

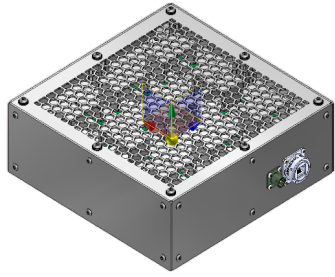
	Including aluminium profiles and mounting brackets: (0.600 x 0.139 x 0.308) m
Experiment expected COG (centre of gravity) position:	Experiment box: Approximately in the center of the box 

Table 4-2: Component list

Component	Quantity	Dimensions [mm]	Mass per piece [g]	Total mass [g]
Exterior Aluminium sheet metal panel	2	254 x 87.5 x 2	119	238
Exterior Aluminium sheet metal panel	2	250 x 87.5 x 2	113	226
20 x 20 Strut profile	4	210 x 20 x 20	94	376
20 x 20 Strut profile	4	82 x 20 x 20	37	148
20 x 20 Strut profile	4	63.5 x 20 x 20	28	112
20 x 20 Strut profile bracket	16	20 x 20 x 18	5	80
Laser-cut mesh-like panel	1	250 x 250 x 2	162	162
Machined Aluminium sample frame	1	250 x 250 x 13	911	884
Machined Aluminium brackets	4	44.5 x 44.5 x 3	9	36
Bottom sheet metal	1	250 x 250	333	333

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panel		x 2		
Aluminium 2017-T6 sample	9	50 x 30 x 30	34	306
Garolite isolating shoes	9	36 x 10 x 8	4	36
Sheet metal coupling	4	42 x 26 x 1.5	15	60
XPS foam	-	-	-	-
M4 T-nut	65	M4	1	73
M6 T-nut	8	M4	1	8
Nordlock® M5 washer	16	M5	1	16
Nordlock® M4 washer	40	M4	1	40
Nordlock® M3 washer	14	M3	1	14
M5 V-coil®	8	M4 x 10	1	8
ISO 7380-1 M5 bolt	4	M6 x 50	7	56
ISO 7380-1 M5 bolt	16	M5 x 16	4	64
ISO 7380-1 M4 bolt	32	M4 x 10	2	64
ISO 7380-1 M4 bolt	24	M4 x 6	1	24
ISO 7380-1 M4 bolt	8	M4 x 12	1	8
ISO 7380-1 M3 bolt	12	M3 x 10	1	12
ISO M3 nut	26	M3	1	26
ISO M5 nut	4	M5	1	4
Square M3 washer	14	M4 x 20 x 1	3	42
30 x 30 Strut profile	2	600 x 45 x 45	500	1000
Profile Bracket R28X28	4	40 x 30 x 28	20	80
Rubber bumpers	8	25 x 12	1	8
M6 T-Bolt	8	M6 x 20	8	64
M6 Flanged nut	8	M6	4	24

Component	Quantity		Mass per piece [g]	Total mass [g]
PT100 Temperature Probe (Platinum Resistance)	10		15	150
MIKROE-2815 RTD to Digital	1		25	25
3.3V 1A STEP-DOWN D24V10F3POLOLU	1		1	1
Logic Stepdown 5v to 3v3	1		2	2
BME280 3V3 module	1		1.5	1.5
BPW 21 Photodiode	1		1	1
Raspberry PI 4 4gb	1		46	46
Pi-EzConnect Terminal Block Breakout HAT	1		34.1	34.1
MICROSD Card 32gb	1		0.5	0.5
Kapton heaters	9		2	18
Polulu 5V 5A buck	1		5	5
Wires, Solder, Nylon standoffs	1		50	50
KKSB ALUMINUM	1		100	100

4.4 Mechanical Design

4.4.1 General Structure Overview

The mechanical structure of the experiment is based on a double-shell configuration, designed to ensure physical separation between the harsh environment and COATHEAL payload. The mechanical assembly is divided into three functional layers:

1. External housing: Sheet metal aluminium panels protect the experiment from debris and contaminants.
2. Internal structure: Strut profile structure that ensures assembly's rigidity and mounting points for ESCARGO.
3. Thermal buffer: Foam layer designed to thermally insulate the electronics from the outside temperatures.

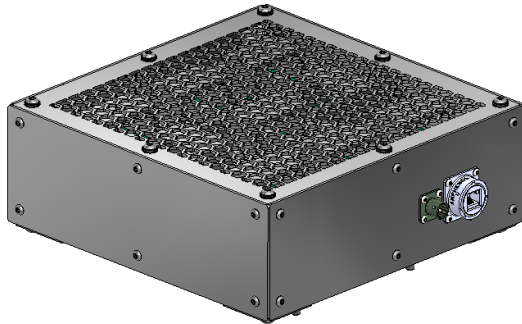


Figure 4.4.1 – Experiment overview

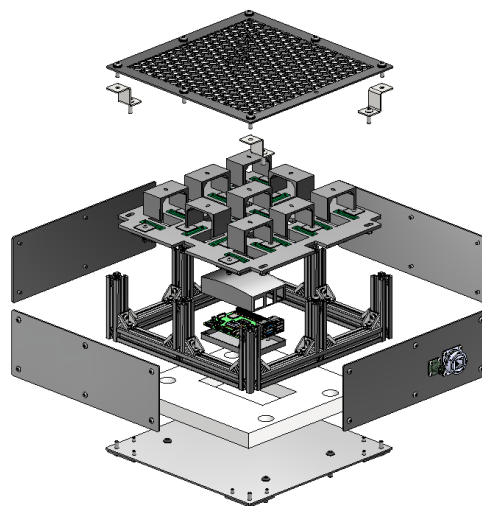


Figure 4.4.2 – Exploded view of the experiment

4.4.2 External housing

The outer shell is composed of five panels fabricated from 2 mm Aluminium 5052-H32 sheet metal, integrated with a mesh-like structure made from the same material. The mesh-like structure was chosen to ensure the absorption of UV radiation as it aids the healing mechanism of the coating. As the mesh-like structure will be laser-cut, no additional machining time is needed compared to using round holes. Compared to the 6061 alloy, 5052 offers superior laser-cutting characteristics, resulting in cleaner edges with minimal

post-processing. As machining would have been expensive for a structural frame, a COTS alternative has been implemented, being based on Bosch Rexroth 20 x 20 aluminium profiles and brackets fastened with M4 bolts and T-nuts, everything being secured by using blue Loctite on the thread and wedge-locking washers (see Figure 4.4.3). We are currently working on implementing an additional support structure on the underside as the bottom aluminium panel could bend as the dimension is big. For each panel, assembly is achieved via six mounting points, everything being fastened into M4 T-nuts inserted in the aluminium profiles, 2 on each channel of a pillar. Upper mesh structure is fastened in the aluminium pillars via threaded inserts with M5 bolts. The same fastening mechanism has been adopted for connecting the box to ESCARGO.

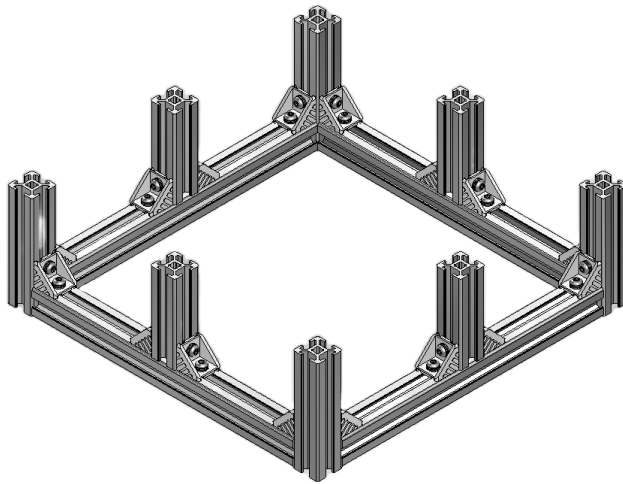


Figure 4.4.3 – Structural frame

In order to ensure structural rigidity for the corners of the frame, laser cut corner reinforcement has been implemented, this system can be seen in Figure 4.4.4. Reinforcements are made out of 3 mm 5052-H32 aluminium alloy with holes for fastening them to the structural frame via M4 bolts with T-nuts.

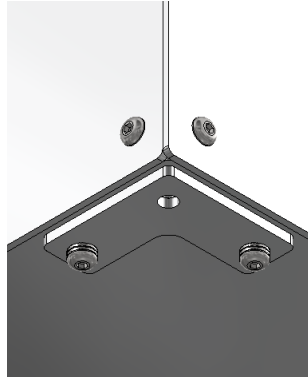


Figure 4.4.4 – Corner reinforcement

4.4.3 Payload

The COATHEAL samples have an updated shape in order to facilitate the appearance of micro-cracks. The system relies on constrained differential thermal expansion and this can be achieved by controlling the temperature of the samples with regards to the temperature of the frame. Having a bigger difference results in higher stresses and thus higher chances of bigger micro-cracks. The experiment aims for $\Delta t \approx 100\text{ }^{\circ}\text{C}$ as this was the simulated value. The samples can be seen in Figure 4.4.5.

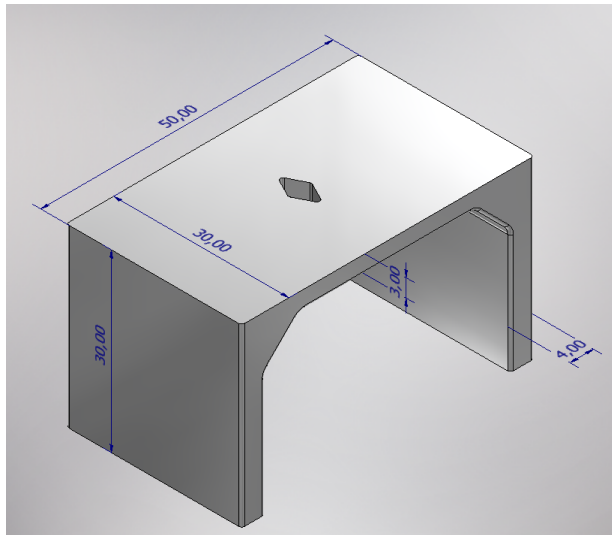


Figure 4.4.5 - COATHEAL sample

The shape of the samples facilitate the generation of stresses through the use of a bigger distance between the applied force location and the surface, thus bending is used as the main damaging action. The diamond shape has been picked as it has the biggest stress concentrating effect we have simulated. Detailed simulations can be found in section 4.4.4 Stress calculation. A thicker sample was also implemented as the thickness generates greater stress at the surface of the sample and ensures catastrophic damage will not occur.

Material was also changed to 2017-T4 aluminium alloy as 3003-H18 would have been very hard to source. This alloy has an elongation at rupture of 22%, thus being ductile enough to ensure a brittle fracture will not appear suddenly but also it is not too ductile thus, the mechanical stress will not result in deflection but in surface micro-cracks. As aluminium 2017 is alloyed with copper, internal stress concentrators appear [1] and this could help the propagation of substrate cracks to the surface. The 2024-T3 alloy was also considered as it has a lower elongation at rupture than 2017-T4 but it could not be sourced in time for tests. Use of 2024-T3 alloy still remains under consideration, if tests demonstrate that 2017 alloy is too ductile, 2024 alloy will be utilized

As micro-cracking is a statistical field, tests will be conducted in the following weeks. The material was sourced already and machining operations have started. The alloy acquired was produced as a cast variant, that could raise the chance of internal stresses thus being beneficial for the experiment. The alloy in this configuration could be considered isotropic, unlike in a rolled state.

The samples will be held in a frame (Figure 4.4.7) constructed from machined 6061-T6 alloy having a total thickness of 13 mm.

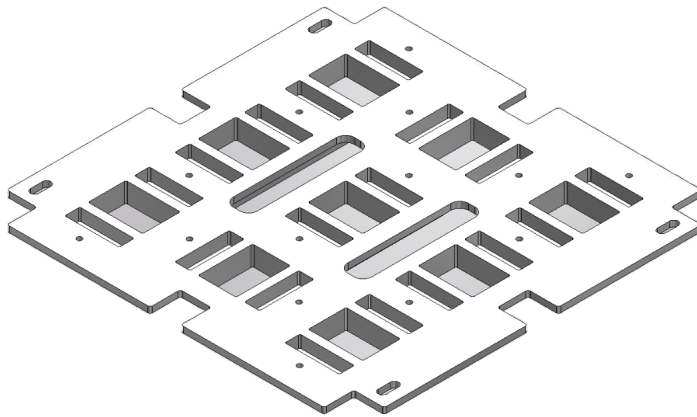


Figure 4.4.6 – Sample frame

Slots are cut in the frame for garolite insulating shoes. Garolite was chosen for its excellent thermal insulation capabilities but also for its hardness and very high tensile strength in the direction of the fibers for a thermoplastic. Garolite will be machined with respect to the fibers so that the orthotropic characteristics will not interfere with the experiment. The garolite spacers will be press-fit into

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the frame and secured with square washers and M3 bolts. The system can be seen in Figure 4.4.8.

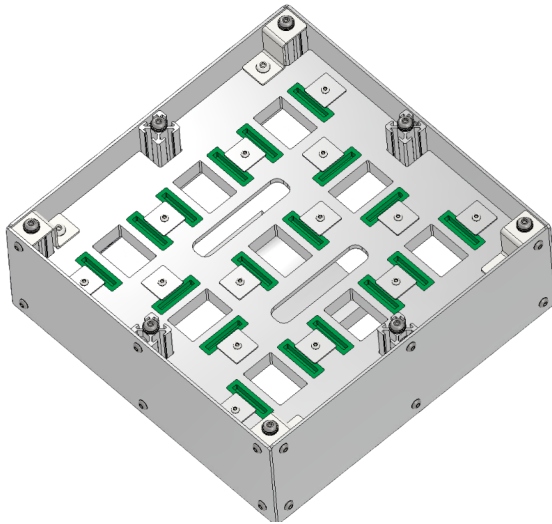


Figure 4.4.7 – Garolite shoes integration

For providing a secure insertion, H7/n6 (Figure 4.4.8) was chosen for a transitional fit with an imperceptible clearance and small interference. The system relies on a hole basis of fits. These fits are detailed in table 4-4.

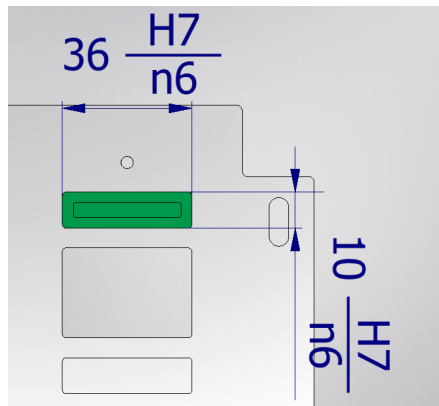


Figure 4.4.8 - Fits for garolite shoes

Table 4-3: Fit and tolerance

Characteristics	Main dimension [mm]	Secondary dimension [mm]
Hole _{min}	36	10
Hole _{max}	36.025	10.015
Hole _{upper}	0.025	0.015
Hole _{lower}	0	0
Shaft _{min}	36.017	10.010
Shaft _{max}	36.033	10.019
Shaft _{upper}	0.033	0.019
Shaft _{lower}	0.017	0.010
Clearance _{max}	0.008	0.005
Interference _{max}	0.033	0.019
Midpoint	-0.0125	-0.007

The sample plate is fastened to the structural frame via four 1.5 mm thick bent sheet metal with M5 bolts with Nordlock® washers. The sheet metal couplings (Figure 4.4.9) are secured to the frame with M5 bolts in threaded inserts in the side pillars.

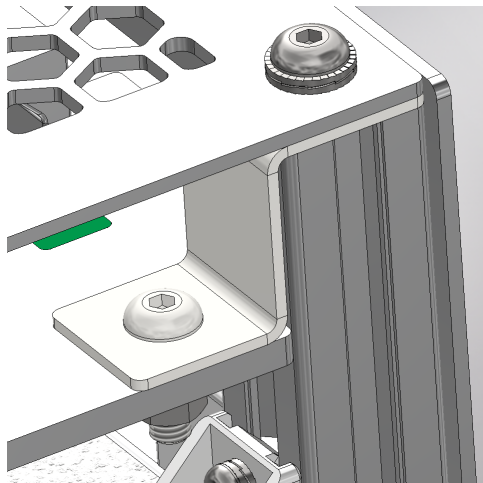


Figure 4.4.9 - Sheet metal coupling

Table 4-4: Supplier list

Component	Quantity	Supplier or manufacturer
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Exterior aluminium sheet metal panels	4	ColorMetalRo, local machining
Mesh-like structure	1	ColorMetalRo, local machining
Machined aluminium sample frame	1	ColorMetalRo, local machining
Machined aluminium brackets	4	ColorMetalRo, local machining
Sheet metal coupling	4	ColorMetalRo, local machining
Garolite isolating shoes	18	CEPROINV, local machining
Bottom sheet metal panel	1	ColorMetalRo, local machining
COATHEAL aluminium samples	9	ColorMetalRO, local machining
XPS foam	600x600x100 sheet	EasyComposites EU
ISO 7380-1 M5 x 60 bolt	100-piece box	TME.EU
ISO 7380-1 M5 x 16 bolt	100-piece box	TME.EU
ISO 7380-1 M4 x 10 bolt	100-piece box	TME.EU
ISO 7380-1 M4 x 6 bolt	100-piece box	TME.EU
ISO 7380-1 M4 x 12 bolt	100-piece box	TME.EU
ISO 7380-1 M3 x 10 bolt	100-piece box	TME.EU
ISO M3 nut	100-piece box	TME.EU
ISO M5 nut	100-piece box	TME.EU
Square M3 washer	50-piece box	HornBach RO
30x30 Rexroth Strut Profile	2 meters	Mig24 RO
Rexroth R Strut profile bracket	8	Mig24 RO
Rexroth 20 x 20 strut profile bracket	16	Mig24 RO
20x20 Rexroth Strut Profile	2 meters	Mig24 RO

M4 T-nut	100-piece box	Mig24 RO
M6 T-nut	100-piece box	Mig24 RO
Nordlock® M5 washer	16	TME.EU
Nordlock® M4 washer	40	TME.EU
Nordlock® M4 washer	14	TME.EU
M5 V-coil®	8	ROCAST
M6 T-bolt	8	TME.EU
M6 Flanged nut	8	TME.EU
Rubber bumper	4	MOUSER ELECTRONICS

4.4.4 Stress calculation

To estimate the mechanical stress, two methods were used: by-hand calculated displacements used as input for FEM analysis and simulating the entire process using FEA.

Regarding the first method, the total displacement shall be modelled as a relative displacement between the sample's fixed supports, in Figure 4.4.8 – left, the right support is considered fixed while the relative displacement is shown on the left support. Having a symmetrical arrangement, the system can be simplified to an L-shaped profile (Figure 4.4.8 – right).

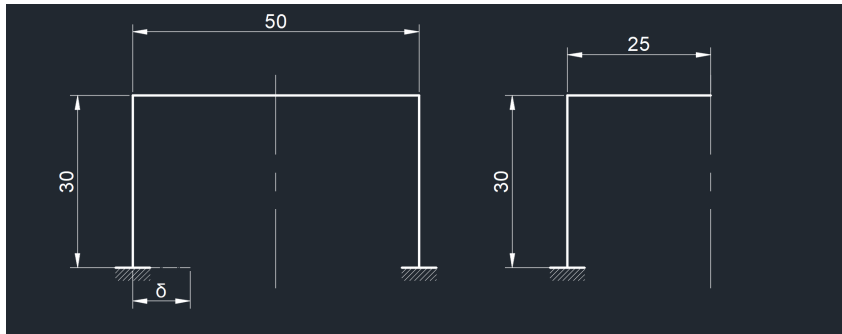


Figure 4.4.10 - Aluminium sample simplified model

The displacement due to thermal effect (δ) was estimated using the relationship between the thermal expansion coefficient of aluminium ($\alpha = 26 \times 10^{-6} K^{-1}$), the object's length ($l = 25 \text{ mm}$) and the temperature difference ($\Delta T = 80 \text{ K}$), as follows:

$$\delta = \alpha \times l \times \Delta T \quad (4.4.3)$$

$$\delta = 0.048 \text{ mm} \quad (4.4.4)$$

The calculated value is to be used as input for FEA for stress calculation. The lower end of the halved sample is modelled using a fixed support and a remote displacement of 0.048 mm was applied on the cantilever end.

The solution yields a horizontal reaction force of 11.3 N, which at a leverage distance of 30 mm results in a bending moment (M_y) of 339 Nmm. The normal stress (σ) at the middle line of the sample can be calculated by the following formula:

$$\sigma = \frac{M_y}{W_y} \quad (4.4.5)$$

$$W_y = \frac{B \times H^2}{6} \quad (4.4.6)$$

where W_y is the flexural modulus, B and H are the base and height of the rectangular cross section. Hence, σ results as:

$$\sigma = \frac{339 \times 6}{30 \times 3^2} = 7.53 \text{ MPa}$$

Due to the absence of stress concentrating effects and the calculus simplifications within the analytical model applied on small-scale systems, real stresses are higher and were checked using FEM analysis in Ansys.

A static structural simulation was conducted in order to find the optimal thickness of the tested sample and the optimal geometric form of the stress concentrator. In the first iteration of the samples, the stress concentrator was a V-shaped notch that had the factor of stress concentrating calculated analytically. After discussions with mechanics of materials professors, we understood that there are too many variables for stress concentrators in order to be calculated using a formula therefore static simulations were conducted. The stress concentrating factor for the V-notch was not satisfactory and iterations of finding a stress concentrator started.

All simulations have the following conditions applied:

- Bonded connections between objects
- Frictionless support on 3 faces as seen in Figure 4.4.11
- $\Delta t \approx 100^\circ\text{C}$ between the frame and samples
- Simplified CAD model according to the variant tested

Meshing conditions:

- Sample: Tetrahedrons, element size of 0.4 mm
- Garolite shoes: Tetrahedrons, element size of 0.5 mm
- Frame: Tetrahedrons, element size of 2 mm

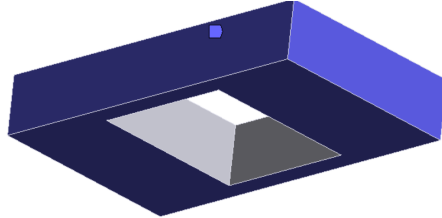


Figure 4.4.11 - Frictionless support position

First iteration included 2 circles of $d = 5$ mm and a thickness of $h = 2$ mm. Results for Equivalent Von-Mises stress [MPa] can be seen in Figure 4.4.12 and Figure 4.4.13. The maximum stress appears in singularities given by the bonded connections and may be ignored as it does not reflect the real behavior.

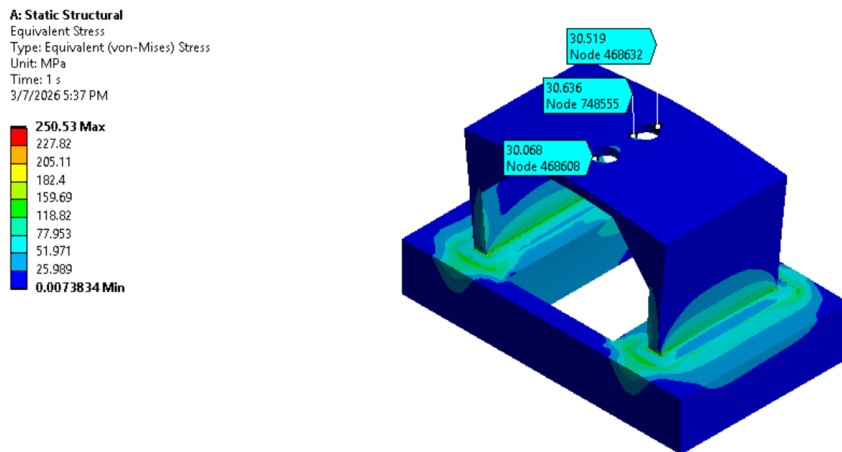


Figure 4.4.12 - First static structural simulation

As it can be seen, the bending stress at the surface is $\sigma \approx 31\text{MPa}$. Given the condition that stress is greater at the extremities of a cross section, the next iteration includes a thickness of $h = 3$ mm and the same round holes as in the first one.

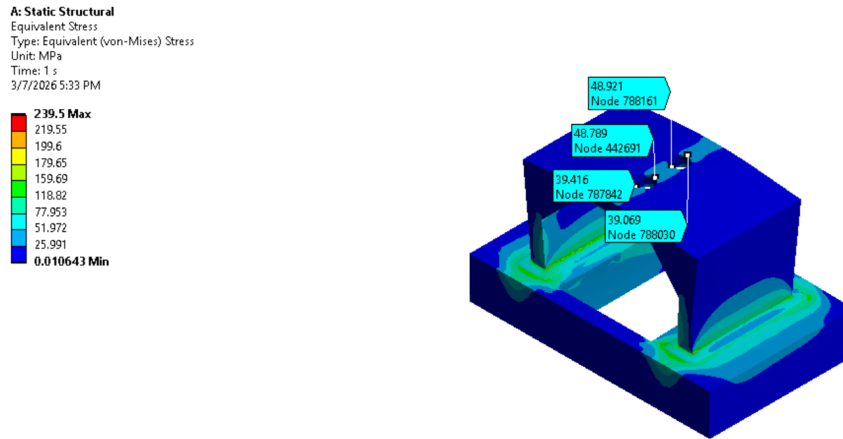


Figure 4.4.13 - Second static structural simulation

The greater thickness increased the surface stress to $\sigma \approx 49\text{MPa}$.

Another version was used for defining the sample having $h = 3\text{ mm}$ and the stress concentrator being diamond shape with the $L = 8\text{ mm}$, $w = 4\text{ mm}$ and machining fillets of $r = 0.5\text{ mm}$.

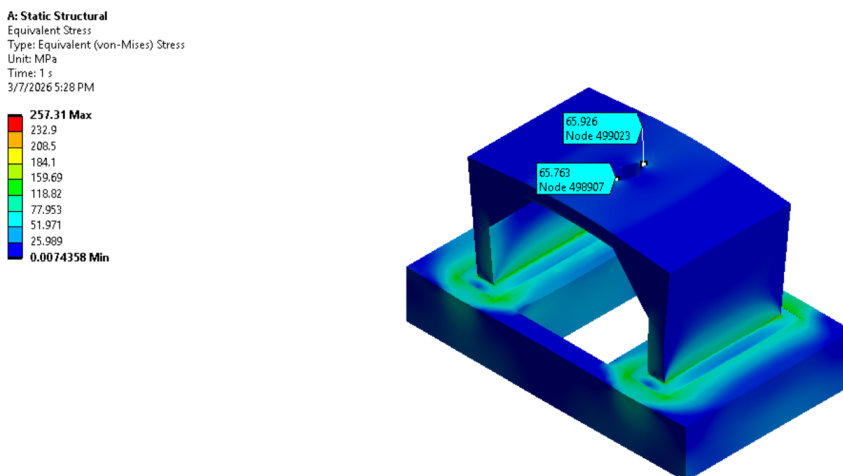


Figure 4.4.14 - Third static structural simulation

Figure 4.4.14 confirms that the diamond shape has a stress concentration factor bigger than those tested before, the stress having a magnitude of $\sigma \approx 66\text{MPa}$. A closer view can be seen in Figure 4.4.15.

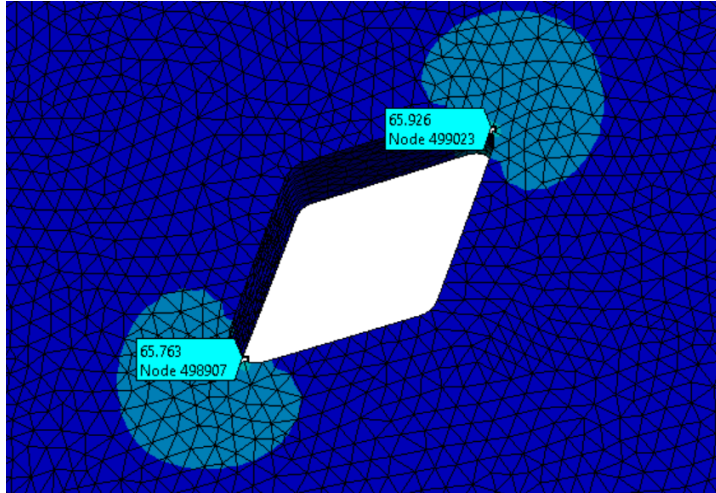


Figure 4.4.15 - Stress concentration

4.5 Electronics Design

4.5.1 Electronics System Overview

The main processing unit in the electronics system is a Raspberry Pi 4 which is suitable for the ambient temperatures reached during the stratospheric flight. Hardware longevity and data integrity will be obtained by keeping the temperature at a constant 0° C floor. Aluminium plates will be controlled from - 30°C hibernation state during ascent to a +70°C for the healing phase. To stay within the recommended 1.4 A (40.32 W) constant draw limit, the system utilizes dynamic power allocation and PWM throttling.

The Control Unit, centered on the Raspberry Pi 4, executes the flight software and manages the logic for all phases of the experiment. Control Unit runs a custom PID control loop that processes real-time data from the sensor array to determine the desired duty cycles for the heaters. Raspberry Pi 4 manages thermal regulation and data logging at high-frequency, storing information from 12 different sensors. By monitoring pressure and altitude data, the controller autonomously triggers the activation of healing phases while it remains accessible via the E-link for emergency overrides.

The Sensor Subsystem performs data acquisition which will be used for real-time and post-flight self-healing process analysis. For temperature measurements, ten PT100 RTD sensors are used, nine mounted under every aluminium plate and one for monitoring ambient temperature. These sensors are interfaced through MIKROE-2815 RTD amplifiers on the SPI bus. For

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ultraviolet radiation measurements, the system has a BPW 21 photodiode. Attitude and pressure are monitored using a BME280 digital barometric sensor.

The Heating Subsystem provides the thermal energy required for the experiment by utilizing nine 10 W Kapton heaters, mounted under every aluminium plate and one 10 W Heater in the Electronics Box. These actuators are controlled via a 9-channel MOSFET driver module that interfaces directly with the Pi's GPIO pins.

This setup allows for fine-grained Pulse Width Modulation, critical requirement for achieving approximately 0.85°C/s heat ramp for activation of the experiment. The software logic ensures that the heaters are prioritized, capping the electronic heater whenever the plates require a maximum ramp-up to stay within the 1.4 A safety envelope.

Power Subsystem manages the distribution of power and serves as the primary interface between the experiment and the 28.8 V DC supply from the BEXUS gondola.

For protecting the 5.1 V electronics from the voltage heater lines and noisy switching signals, a Pololu D24V50F5 buck converter is used. This regulator steps down the raw voltage to a stable 5.1 V at 5 A, providing clean power to the Pi4. This architecture ensures that power fluctuations caused by high-current heater pulses do not interfere with the stability of the control unit or sensor readings.

The Communication Subsystem facilitates real-time telemetry and ground control via the BEXUS E-link system. An Ethernet connection provides the physical link between the Raspberry Pi 4 and the gondolas communication hub.

During flight, the experiment transmits a continuous telemetry packets, including sensors measurements, heaters states and power, allowing the ground support team to monitor mission phases. The uplink capability also provides remote shell access via TCP, ensuring that the team can perform diagnostics or reset the system if the autonomous flight logic unforeseen atmospheric conditions.

4.5.2 Electronics Architecture and Logic

The electronics architecture separates low-power logic from high-power logic from high-power actuation to prevent noise from interfering with sensor readings. A processing unit coordinates all systems and manages data flow and thermal regulation. The experiment electronics are organized into a central logic that interfaces with power, sensing, and thermal control subsystems.

The following block diagram shows the overall system architecture, highlighting the separation between the 28.8V power rails and the 5.1V logic signals.

As shown in Figure 4.5.1, the primary power line has two distinct purposes. Firstly, it feeds into a Buck Converter which steps down the voltage value to a stable 5.1 V to power the Raspberry Pi 4 and the sensors subsystem. Secondly, the 28.8V line is routed directly to the Heater Control Path.

Heater Control Path

The heating subsystem maintains the electronics box at a survival temperature and helps the plates to reach their 70°C target temperature for the healing process. This path uses a 9-Channel MOSFET Driver that functions as an array of high-speed electronic switches. MOSFETs are connected to the 28.8 V power rail and controlled by the Raspberry Pi 4 via Pulse Width Modulation (PWM) signals.

PWM use allows the control software to rapidly toggle the MOSFETs on and off, effectively “throttling” the power delivered to the nine 10W Plate Heaters and the 10W Electronics Box Heater.

The Raspberry Pi continuously calculates optimal power percentage for each heater based on real-time feedback from the temperature sensors and adjusts the duty cycle of the PWM signals to maintain the desired thermal profile.

Data Flow and Acquisition

The data flow is a multi-stage process that sends environmental data to ground-based stations for monitoring. The Sensor Subsystem collects data across several digital buses. Data is processed by the Raspberry Pi.

The I2C Bus is used to poll the BME280 digital barometric sensor and the ADC, which transforms the analog signal from the BPW 21 UV photodiode.

The SPI Bus manages 10 MIKROE-2815 RTD amplifiers, which read the values of the electrical resistance from the 10 PT100 RTD sensors located on the plates and inside the chamber.

When the Raspberry Pi 4 receives the data, the flight software manages it in two ways. Internally, the data is immediately timestamped and written on an Industrial MicroSD card to ensure a persistent, non-volatile record of the observed data during flight.

Simultaneously, the Pi splits this data into telemetry frames and transmits them over Ethernet to the BEXUS E-Link.

4.5.3 Subsystem Description

The design utilizes the Raspberry Pi 4 as the primary controller for the COATHEAL experiment. Because the system requires a full operating system to manage multiple high-speed data buses simultaneously, the Pi acts as the central hub, polling sensors, executing the thermal control loops for polymer healing, and handling the telemetry stream. The processing overhead supports a continuous flight software loop that performs real-time calculations while

guaranteeing all collected data is mirrored redundantly to both a local microSD card and a USB stick to prevent data loss.

For the heater control subsystem, the architecture centers around nine 10 W Kapton heaters dedicated to the sample plates driven by MOSFET switching modules. The software implements an active/queued Time-Division Multiplexing (TDM) strategy combined with Pulse Width Modulation (PWM). The PID control loop relies on exact feedback from ten PT100 RTD sensors. The Pi handles these readings and outputs a 3.3 V PWM signal to the MOSFETs, which dynamically switch the high-voltage 28.8 V line to the active heaters, safely and reliably retaining the 70°C target during the healing phase.

The sensor subsystem provides comprehensive mapping of the stratospheric environment and the experiment's internal thermal state. To prevent data congestion, the architecture splits the sensor array among distinct electronic interfaces. The system utilizes the SPI bus for the ten MIKROE-2815 RTD amplifiers, providing the high speed and individual chip-select lines necessary for rapid temperature measuring across all plates. For ambient environmental data, the I2C bus supports the BME280 sensor to track pressure, humidity, and external temperature. Additionally, the I2C bus interfaces with an ADC, which processes the analog output from the BPW 21 UV photodiode and delivers the high resolution required for radiation analysis.

Regarding the communication subsystem, the experiment interfaces with the BEXUS E-link through a dedicated Ethernet connection. The data flow prioritizes live telemetry over a reliable TCP/IP protocol, utilizing newline-delimited records to transmit continuous sensor and system health packets to the ground station. While the downlink supports high-bandwidth transmission of the complete sensor array data, the uplink is programmed to accept essential commands, such as manually overriding mission phases, forcing a heater shutdown, or resetting the control unit. The packet structures and transmission frequencies are strictly optimized to ensure uninterrupted monitoring without saturating the allocated E-link bandwidth during flight.

4.6 Thermal Design

4.6.1 Thermal context

During the experiment's ascent, temperatures may drop down to -80 °C within the tropopause. Convection is directly linked to air pressure, resulting in a fast reduction of the film coefficient. If a daytime flight is considered, the date and time, as well as geographic coordinates must be taken into account to estimate the absorbed solar flux. A nighttime flight shall be considered as well, due to easier activation of the cracking mechanism, as greater temperature differences may be achieved.

4.6.2 Physical arrangement

The described conditions pose a significant threat for the electronics, hence thermal insulation (extruded polystyrene – XPS) will be used to protect the Raspberry Pi.

For thermal insulation between the COATHEAL samples and the sample frame, garolite (FR4) shoes will be used. The orthotropic thermal conductivity is accounted for in the numerical simulation.

4.6.3 Analysis setup

A simplified geometry (Figure 4.6.1), which respects the key features of the original model was imported into Ansys to study the thermal behaviour of the experiment.

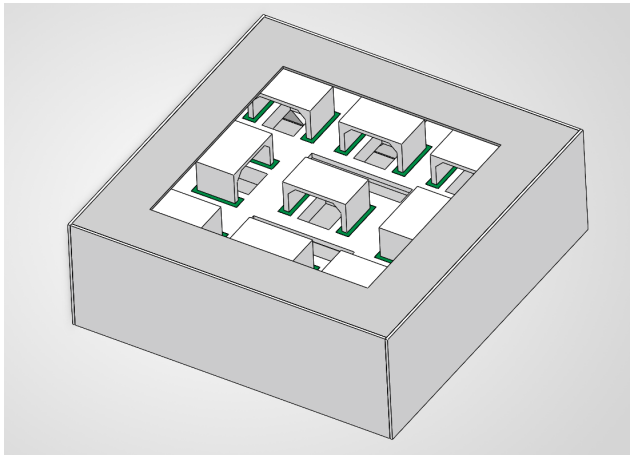


Figure 4.6.1 – Simplified 3D model

The thermal properties (thermal conductivity and specific heat) of the aluminium alloys used were defined with respect to temperature using approximation curves determined by the following formula:

$$y = 10^{a+b(\log_{10}T)+c(\log_{10}T)^2+d(\log_{10}T)^3+e(\log_{10}T)^4+f(\log_{10}T)^5+g(\log_{10}T)^6+h(\log_{10}T)^7+i(\log_{10}T)^8} \quad (4.6.1)$$

where:

- y – the calculated thermal property (e.g. the thermal conductivity of aluminium 6061-T6).
- a ... i – coefficients for curve approximation.
- T – the temperature (in Kelvin) at which the thermal property is calculated.

Calculated results are shown in Table 4-4, along with coefficients used for each calculation.

Table 4-4: Thermal conductivity / Specific heat vs. Temperature

Aluminium 6061-T6			Aluminium 3003			Stainless steel AISI304		
Thermal conductivity		Specific heat	Thermal conductivity		Specific heat	Thermal conductivity		Specific heat
UNITS	W/(m-K)	J/(kg-K)	UNITS	W/(m-K)	J/(kg-K)	UNITS	W/(m-K)	J/(kg-K)
a	0.07918	46.64670	a	0.63736	46.6467	a	-1.4087	22.0061
b	1.09570	-314.29200	b	-1.1437	-314.292	b	1.3982	-127.5528
c	-0.07277	866.66200	c	7.4624	866.662	c	0.2543	303.647
d	0.08084	-1298.30000	d	-12.6905	-1298.3	d	-0.626	-381.0098
e	0.02803	1162.27000	e	11.9165	1162.27	e	0.2334	274.0328
f	-0.09464	-637.79500	f	-6.18721	-637.795	f	0.4256	-112.9212
g	0.04179	210.35100	g	1.63939	210.351	g	-0.4658	24.7593
h	-0.00571	-38.30940	h	-0.172667	-38.3094	h	0.165	-2.239153
i	0.00000	2.96344	i	0.00000	2.96344	i	-0.01990	0
Temperature	Thermal conductivity	Specific heat	Temperature	Thermal conductivity	Specific heat	Temperature	Thermal conductivity	Specific heat
Celsius Kelvin			Celsius Kelvin			Celsius Kelvin		
0 273.15	151.31791	917.97703	0 273.15	175.56038	917.97703	0 273.15	14.59687	468.46919
-5 268.15	150.48968	912.69466	-5 268.15	175.03505	912.69466	-5 268.15	14.46407	466.85804
-10 263.15	149.63353	907.65889	-10 263.15	174.47806	907.65889	-10 263.15	14.33119	464.85622
-30 243.15	145.91334	888.49296	-30 243.15	171.96942	888.49296	-30 243.15	13.79859	453.45657
-40 233.15	143.86299	878.48841	-40 233.15	170.57252	878.48841	-40 233.15	13.53124	446.08188
-50 223.15	141.67474	867.46641	-50 223.15	169.10053	867.46641	-50 223.15	13.26269	437.85263
-52 221.15	141.21973	865.09360	-52 221.15	168.79843	865.09360	-52 221.15	13.20878	436.11998
-54 219.15	140.75875	862.65518	-54 219.15	168.49403	862.65518	-54 219.15	13.15480	434.36147
-56 217.15	140.29170	860.14718	-56 217.15	168.18746	860.14718	-56 217.15	13.10074	432.57830
-58 215.15	139.81852	857.56571	-58 215.15	167.87882	857.56571	-58 215.15	13.04658	430.77166
-60 213.15	139.33911	854.90693	-60 213.15	167.56822	854.90693	-60 213.15	12.99233	428.94264
-62 211.15	138.85354	852.13313	-62 211.15	167.25562	852.13313	-62 211.15	12.93798	427.08674
-65 208.15	138.11273	847.89721	-65 208.15	166.78391	847.89721	-65 208.15	12.85624	424.27902
-70 203.15	136.84539	840.32397	-70 203.15	165.98975	840.32397	-70 203.15	12.71938	419.49674
-75 198.15	135.53554	832.13313	-75 198.15	165.18732	832.13313	-75 198.15	12.58160	414.60782
-80 193.15	134.18153	823.27376	-80 193.15	164.37808	823.27376	-80 193.15	12.44276	409.62119

The resulting curves were plotted and are shown in Figure 4.6.2.

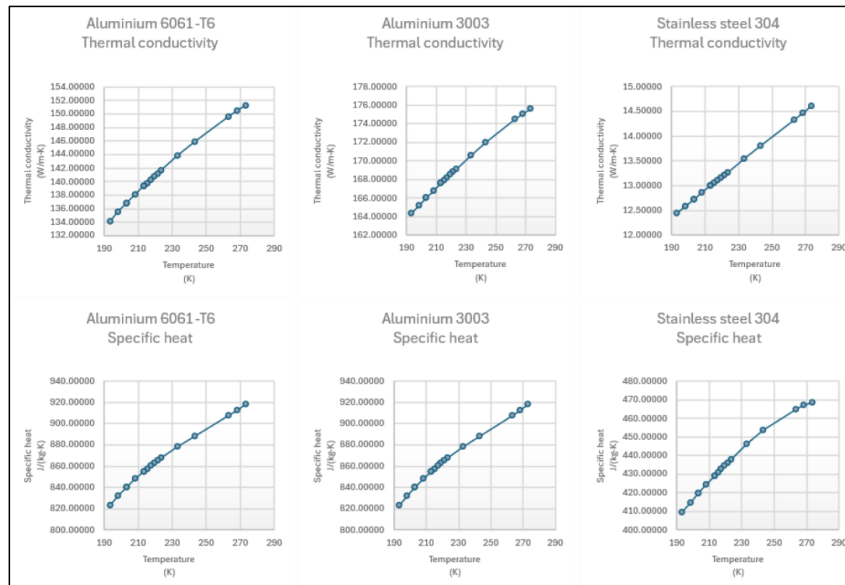


Figure 4.6.2 – Thermal conductivity / Specific heat curves with respect to Temperature

For the remaining materials (XPS, FR4), no detailed data was found and thus, were assumed to have a linear relationship between thermal conductivity / specific heat and temperature.

Material assignments:

- Aluminium 6061-T6: Rexroth profiles, sheet metal panels, sample frame, Raspberry Pi case.
- Aluminium 3003: samples (even though we will use aluminium alloy 2017, thermal characteristics are similar between the two alloys);
- Stainless steel 304: Z-shaped links connecting the sample frame to the Rexroth profiles.
- XPS: thermal insulation around Raspberry Pi.
- FR4: insulating shoes on the samples.

Two separate numerical simulations were run, representing conditions for both daytime and nighttime flight conditions.

To accurately model the absorbed solar flux during a daytime flight, sun positioning data was retrieved using SunCalc, a free-to-use online solar calculator. Having an altitude of $\sim 14^\circ$ to the horizon, the Sun will mostly heat the lateral panels of the experiment. The solar radiation that hits the top of the experiment will be:

$$G_{top} = G \times \sin(14^\circ) \quad (4.6.2)$$

In order to calculate lateral solar flux, constant rotation of the gondola was assumed. The average flux is equal to the integral mean value of the instantaneous flux, divided by the total period (2π):

$$G_{avg,lateral} = \frac{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} G \times \cos(14^\circ) \times \cos(\theta) d\theta}{2\pi} \quad (4.6.3)$$

where θ is the gondola rotation angle, relative to the Sun. At ground level, G is set to 650 W/m^2 , gradually increasing to 1360 W/m^2 at float altitude. These calculations yield the values shown below in Table 4-5, after taking into consideration the absorptivity for each material.

Table 4-5 – Solar flux for all surfaces

AI solar flux TOP	AI solar flux LAT	COATHEAL solar flux TOP	AI solar flux MATRIX
55.04	70.26	49.53	27.52
55.04	70.26	49.53	27.52
63.50	81.07	57.15	31.75
71.97	91.88	64.77	35.99
84.67	108.10	76.21	42.34
97.37	124.31	87.64	48.69
105.84	135.12	95.26	52.92
108.38	138.37	97.54	54.19
110.92	141.61	99.83	55.46
111.77	142.69	100.59	55.88
112.61	143.77	101.35	56.31
113.46	144.85	102.12	56.73
114.31	145.93	102.88	57.15
114.73	146.47	103.26	57.37
114.73	146.47	103.26	57.37
115.15	147.01	103.64	57.58
115.15	147.01	103.64	57.58
115.15	147.01	103.64	57.58

Boundary conditions (daytime flight):

- Internal heat generation, 5W uniformly distributed throughout the Raspberry Pi case body.
- Convection 1, applied onto the external surfaces (detailed in Figure 4.6.3).
- Convection 2, applied to surfaces under the upper grill, having a halved film coefficient.
- Radiation to ambient, on the COATHEAL samples ($\epsilon = 0.9$), the sample frame and the outer panels ($\epsilon = 0.15$), where ϵ is the emissivity of each material.
- Heat flux, simulating the solar flux, applied as specified in Table 4-5.
- Heat flux, representing the heaters on the samples, 10W distributed onto the lower surface of each sample, activated one by one with some overlapping.

The boundary conditions for the nighttime flight simulation only differ by the absence of solar flux, all other conditions remain unchanged.

		Ambient temp.	Convection 1	Convection 2
Tabular Data				
Steps	Time [s]	Temperature [°C]	Convection Coefficient [W/m ² ·°C]	Convection Coefficient [W/m ² ·°C]
1	1	0.	10.	5.
2	1	7200.	10.	5.
3	1	7800.	6.7	3.35
4	1	8400.	4.5	2.25
5	1	10800	1.3	0.65
6	1	11500	0.9	0.45
7	1	12500	0.5	0.25
8	1	12800	0.45	0.225
9	1	13100	0.4	0.2
10	1	13400	0.35	0.175
11	1	13700	0.3	0.15
12	1	14000	0.25	0.125
13	1	14300	0.2	0.1
14	1	14600	0.2	0.1
15	1	14800	0.15	0.
16	1	15000	0.	0.
17	1	25000	0.	0.
18	1	28800	0.	0.

Figure 4.6.3 – Convection related tabular data in Ansys

Initial temperature is set to 20 °C on the entire model to simulate pre-launch conditions.

Time stepping:

- Total time – 8 hours (28800 s).
- Initial time step – 1500 s.
- Maximum time step – 3000 s.
- Minimum time step – 150 s.

Meshing was done using 1 mm elements for the samples and FR4 shoes and 4 mm elements for the other bodies.

4.6.4 Results. Interpretation

After solving the system, the solution reveals the following results (Figure 4.6.4 and Figure 4.6.5):

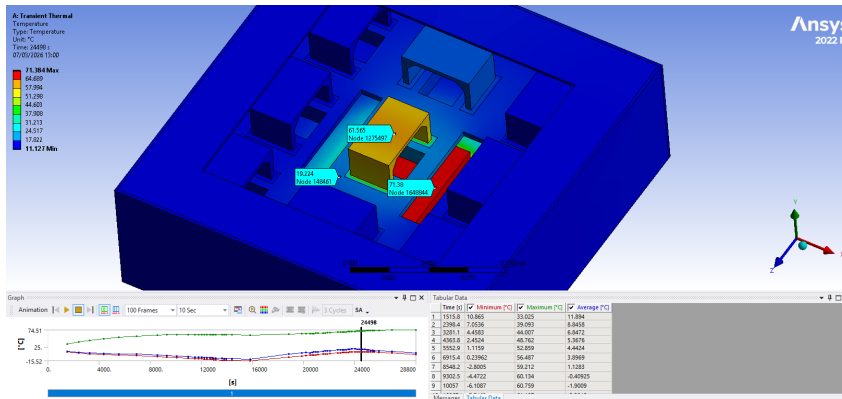


Figure 4.6.4 – Nighttime flight FEA results

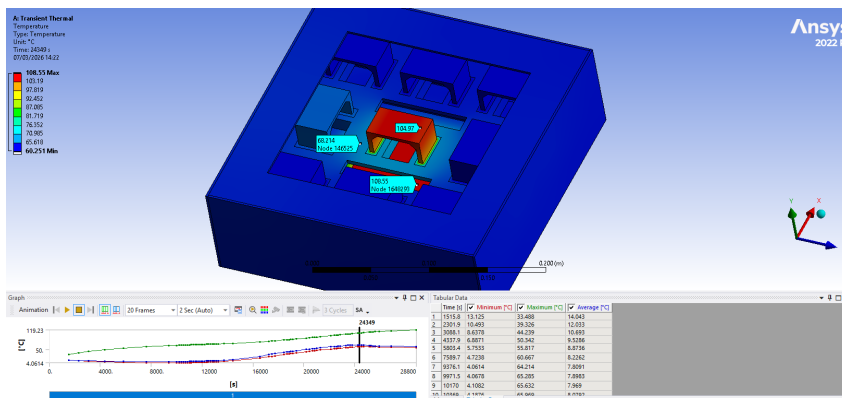


Figure 4.6.5 – Daytime flight FEA results

Between the COATHEAL samples and the frame, in both cases, the temperature difference is ~40 °C. However, a daytime flight will massively increase temperatures, especially after convection becomes negligible, pushing the Raspberry Pi over the temperature limit. Therefore, flying during the night is being considered, further investigation shall reveal a clear answer in future SED versions.

4.7 Power System

4.7.1 Power Supply Overview

The BEXUS 28.8 V DC bus powers the experiment. With a total energy need of 223.9 Wh, our current analysis suggests that using two batteries on the gondola may be needed to provide enough energy and safely handle the peak current draw of 1.73 A.

The electrical system uses a simple power distribution setup to safely manage control and sensor electronics. A DC-DC buck converter creates a stable 5V supply for the Raspberry Pi 4 and the Pi-EzConnect HAT. The sensor array, which includes PT100 probes connected through MIKROE-2815 RTD amplifiers, as well as BME280 and BPW21 sensors, runs 3.3 V logic signals from this supply. The low-power logic circuits include filters to cut down on electromagnetic noise and voltage changes during operation.

The safety plan focuses on real-time software monitoring and several layers of hardware protection. To lower the risk of electrical failures, the system uses fast-acting fuses and reverse-polarity protection at the main power inlet.

4.7.2 Power Budget Calculation

The table presents the current estimation of power usage during flight, incorporating the continuous theoretical draw of the computing and sensor subsystems as well as the dynamically throttled Kapton heaters. Although these figures are subject to refinement during physical testing and integration, the preliminary budget demonstrates that a two-battery configuration can reliably support the calculated 223.9 Wh total load over the anticipated flight duration.

	A	B	C	D	E	F	G	H	I
	Subsystem	Component	Quantity	Voltage	Current (A)	Peak power	Duty cycle estimate	Average power	Total usage (Wh)
1	COMPUTING	Raspberry Pi 4 (4GB)	1	5.0 V	0.22 A	6.00 W	100%	6.00 W	48.0 Wh
2		Pi-EzConnect HAT	1	5.0 V	<0.01 A	0.05 W	100%	0.05 W	0.4 Wh
3	SENSORS	MIKROE-2815 RTD Amp	10	3.3 V	0.01 A	0.25 W	100%	0.25 W	2.0 Wh
4		PT100 Probes	10	3.3 V	<0.01 A	0.01 W	100%	0.01 W	0.1 Wh
5		BME280 / BPW 21	1+1	3.3 V	<0.01 A	0.02 W	100%	0.02 W	0.2 Wh
6	POWER MNGMT	DC-DC Converter	1	28 V	0.05 A	1.40 W	100%	1.40 W	11.2 Wh
7		MOSFET Switching Loss	10	28 V	0.02 A	0.50 W	50%	0.25 W	2.0 Wh
8	THERMAL	10W Heaters (Active)	4	28 V	1.43 A	40.0 W	20%	8.00 W	64.0 Wh
9		10W Heaters (Queued)	5	28 V	0.00 A	0.0 W	20%	12.00 W	96.0 Wh
10									
11	TOTALS				1.73 A (Peak)	48.23 W		27.98 W	223.9 Wh
12									NEEDS 2 BATTERIES
13									

4.7.3 Duty Cycles and Operational Phases

The experiment uses nine 10-watt Kapton heaters. They are placed under the sample plates. The thermal control program manages the power budget as a shared resource and uses time-division multiplexing. Running all nine heaters at once would need about 100 watts, which is more than the system's power limit and the 223.9 watt-hour battery capacity. The software makes sure that no more than four heaters are on at the same time, so the thermal subsystem never draws more than 40 watts (1.43 A), and the system's peak stays at 48.23

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Field Code Changed

watts (1.73 A). The other six heaters wait in a queue and do not draw any current. By quickly switching which heaters are active using Pulse Width Modulation (PWM), the system keeps the right average duty cycles for all temperature zones without going over the power limit.

During ascent, the main goal is to keep the self-healing coating inactive by holding the aluminum plates at -30°C. The plates get short bursts of heat to stop the samples from reaching very low temperatures. The most important change happens during the activation phase, when the controller heats the plates from -30°C up to the +70°C healing target at about 0.85°C per second. At this point, the multiplexer uses its full 48.23 W system peak, focusing on the nine sample plates to reach the needed temperature quickly. To stay within the four-heater limit, the electronics heater is temporarily set aside. The thermal mass of the extruded polystyrene (XPS) insulated box helps keep the inside temperature steady during this fast heating period.

When the experiment reaches the float phase in the stratosphere, most heat is lost through radiation instead of convection. The multiplexing queue becomes less demanding as the system keeps the plates at +70°C for the 90-minute healing period, making sure energy use stays within the two-battery limit. During descent and recovery, the balloon moves into denser air, which increases cooling by convection. The thermal targets then change to a -20°C floor for the plates to stop the newly healed polymer from getting brittle. The system remains running so it can log data during landing and while waiting for the recovery team.

4.8 Software Design

1. Purpose:

The On-Board software covers the following functionalities:

- Temperature regulation is based on the control of the on-board heaters and the measurements read by the temperature sensors.
- Sensor reading and management for the ambient (temperature/pressure/humidity), temperature (heaters) and UV sensors
- Data storage in a designated log file with timestamps
- Data transmission to the Ground Station.
- Receiving and enacting commands from the Ground Station

The Ground Station software covers the following functionalities:

- Parsing of the collected data received from the gondola
- Computing relevant data
- Displaying the data in a readable graph
- Transmitting commands to the gondola

2. Design:

a. Process Overview:

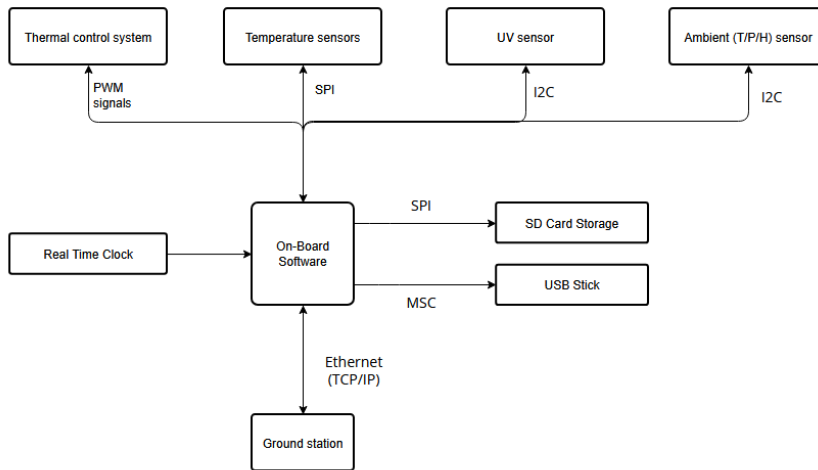


Figure 4.8.1 – Overview Block Diagram

As displayed in figure 4.8.1, the On-Board Computer communicates with UV and ambient sensors through I2C connections, with the temperature sensors through SPI and with the thermal control system using PWM signals. An external Real Time Clock provides the means for timestamping our data. The SD Card and the USB Stick are connected through SPI and MSC respectively. Finally, the Ground Station and On-Board computer exchange data through E-link.

b. General and safety related concepts

For a failsafe system against data loss, the collected data is stored twice, locally, on 1 microSD card and 1 USB Stick, and sent in real-time and stored on the Ground Station. Telemetry and commands are transported over a TCP/IP connection via E-Link (Ethernet). TCP provides ordered, reliable delivery of bytes; message boundaries are defined at the application layer using newline-delimited records.

c. Interfaces

Table 4-6: Interface table

Components		Communication Protocol	Physical Interface	
A	B		B	A
OBC	TS	SPI	Digital Pins	GPIO pins
OBC	UVS	I ² C	Digital Pins	GPIO pins
OBC	AS	I ² C	Digital Pins	GPIO pins
OBC	TCS	PWM	Digital Pins	GPIO pins using MOSFET driver
OBC	SD card	SPI	SD card edge connector	Micro SD slot
OBC	USB Stick	MSC	USB-A plug	USB-A port
OBC	GS	E-Link	Ethernet	Ethernet

Table 4.81 displays all component communications and their respective communication interfaces and physical connections. The UV and ambient sensors communicate through I²C with the On-Board Computer while the temperature sensors and the temperature control system communicate through SPI and PWM. The SD card and the USB Stick communicate with the On-Board Computer through SPI and MSC respectively. Finally, the On-Board Computer sends collected data and receives commands through E-Link.

d. Data acquisition and storage

- Data will be read from the temperature, ambient, and UV sensors at a specific tick-rate and stored in packets. For each tick, a packet will contain the timestamp, and all sensor readings in a newline-delimited CSV. Example formatting:
 SEQ (monotonic counter)
 TIMESTAMP (RTC time + "RTC valid" flag)
 T_ambient, P_ambient, UV, T_box (etc.)
 HEATER_DUTY
 STATUS bitfield (SD_OK, USB_OK, I2C_OK, SPI_OK, LINK_OK)

e. Process Flow:

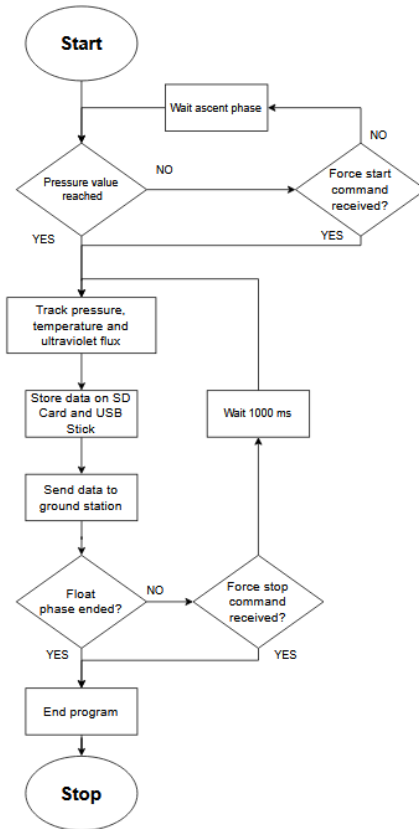


Figure 4.8.2 – Data collection flow diagram

The On-Board Software serves 2 main purposes, each having their own individual process flows. The first functionality is monitoring and collecting data. The collected data is further stored on the SD Card and on the USB Stick, being then sent to the Ground Station to be used and displayed. Before starting to collect data, the thread first checks if the ascent phase is over, determined by the currently measured pressure value. The same criteria can be used to assert if the floating phase is over. The experiment can also be assisted in detecting the current phase through manual commands, such as the force start command or the force stop command.

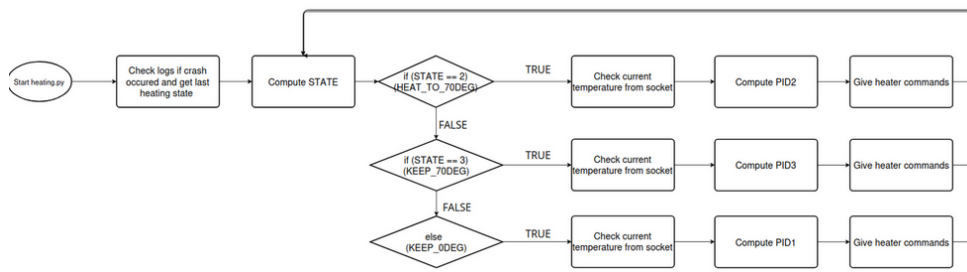


Figure 4.8.3 – Heater regulation flow diagram

The second functionality is responsible for automatically controlling the heaters to maintain an optimal temperature for each sample. While in the ascent phase, the heaters will heat up the samples to 70 degrees. During the floating phase, depending on their current temperature, the heaters will either reduce or increase their output in order to maintain constant temperature. During the descent phase, the heaters will stop.

4.9 Ground Support Equipment

The Ground Station includes 2 laptops, one running the experiment, and one spare laptop for redundancy. The laptop running the experiment will be connected to the E-Net network to access the E-Link and communicate with the On-Board Computer. The uplink will be used to send emergency commands to the On-Board Computer, and the downlink will be used to receive relevant data from the sensors on board, both limited to a bandwidth of 2Mbps to adhere to the imposed hardware constraints.

On the running laptop, team members will be able to monitor relevant data received or calculated based on values received from the On-board Computer. Furthermore, commands such as ON, OFF, or RESET can be issued to the On-Board Computer using the uplink.

4.10 Coating Chemistry

4.10.1 State of the Art in Self-Healing Coatings

Self-healing materials are synthetically-created or fluid artificially made substances which through chemical and physical reactions, have the inert ability to repair the affected regions, without external interventions [2], [3].

Self-healing composites have been placed in a class of systems that could repair their structure either autonomously, extrinsic, or with external aid, intrinsic [3].

Extrinsic Self-Healing

Significant research on autonomic self-healing polymers has focused on crack-filling mechanisms. These healing agents could consist of liquid monomers, catalysts, thermoplastic polymers, film-forming materials or solvent solutions [2].

The main advantage of this technique is that it does not require the damage surface to come back into contact and exactly restructure the compound, especially due the polymer rigidity [2].

Although, the mechanism could crack-fill, only small areas, this approach is more suitable for addressing a wider range of damaged sizes. This self-healing system is more effective when the loading of healing agent is sufficient for filling the entire cracked surface [2].

One of the most researched system for self-healing functionality involves the incorporation of microcapsules containing a liquid monomer to be the healing agent and a catalyst fixed into the polymer matrix [2], [3], or in additional microcapsules [3]. The microcapsules rupture at the propagation of the material composite, causing the liquid monomer flow into the crack surfaces and upon contact with the catalyst, the crack heals [2], [3]. The healing results from the polymerization of the healing agents brought into contact with the catalyst [2], [3].

This capsule-based self-healing system has been thoroughly investigated as it presents a wide range of applications, including coating industries, adhesive industries, and paint industries.

One system was done using polyurethane microcapsules containing hexamethylene diisocyanate as a core material in polymeric composites, enhancing the corrosion resistance of coatings [3]. The synthesis includes the reaction of a diol-diisocyanate prepolymer based on 1,4-butanediol and toluene-1,4-diisocyanate or 4,4-diphenylmethane diisocyanate to prepare polyurethane PU microcapsule shell [3].

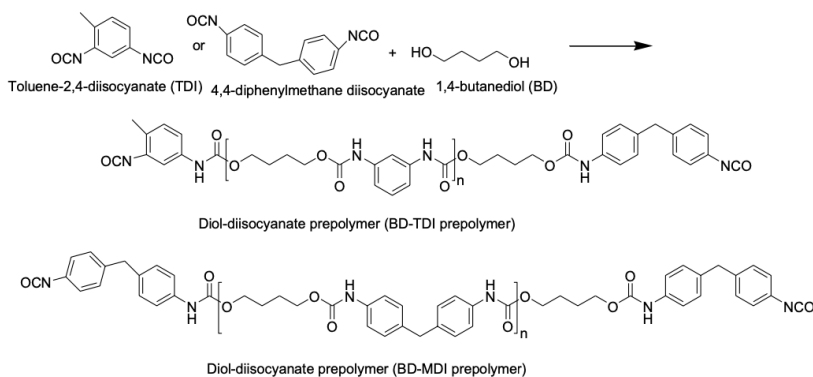


Figure 4.10.1 - Synthesis of PU microcapsules using a prepolymer (BD-TDI prepolymer) and a chain extender (BD) [3]

Intrinsic Nanocomposites Self-healing

The concept of self-healing nanocomposites was inspired by spontaneous repairable biological systems [2], [3]. Due to lower mechanical performance of the polymer composites, they are less prone to show degradation such as micro-cracks, on the material surface, comparing to metals and ceramics [3]. Such that, the introduction of nanocomposites have demonstrated to significantly improve the physiscal and mechanical properties of polymers. The different phases could not individually achieve these properties , instead the nanomaterials are embedded into the host matrix yields material [3].

Various applications were obtained due to the incorporation of nanofillers or nanomaterials, such as carbon nanotubes, graphene, silica, nanoclays, poly-oligomeric silsesquioxanes [3].

One of the highest development of self-healing nanocomposites is represented by carbon-based nanomaterials, such as nanotubes and graphene, due to their mechanical strength, special thermal properties and electrical conductivity [3]. They are used as nanoreserviors for the healing agents when confined within single-walled carbon nanotubes, allowing the fabrication of self-healing materials [3].

Similar, polyhedral oligomeric silsesquioxanes (POSS) are used as building blocks for the production of organic-inorganic hybrid polymer materials [3]. The attractive aspect, related to self-healing system, is their thermal stability, mechanical performance and the ease of the incorporation of the desired functional groups into the side chain [3].

Nanoclays are extermelly important in the production of self-healing composites not only because of their exfoliated arrangements in the polymer, but also

because of their flame-retardant property [3]. This clay-based self-healing has found to be successful in the field of aeronautic engineering, due to these properties that increase the electrical conductivity [3].

4.10.2 Selected Healing Chemistry

Healing agent

On the basis of a thorough literature analysis, dicyclopentadiene DCPD was selected as the optimal choice as a healing agent.

It could be easily associated with 5-ethylidene-2-norbornene known as ENB, due to their utilization in the ring-opening metathesis polymerization mechanism, as part of self-healing systems.

Given this similarity, a comparison between poly-urea-formaldehyde(PUF)-dicyclopentadiene (DCPD) and melamine-urea-formaldehyde (MUF)-5-ethylidene-2-norbornene (ENB) systems was performed in order to determine the suitability of the DCPD monomer.

ENB, a colorless liquid compound is known for its greater ring resistivity and a higher reactivity compared to DCPD. During the laboratory procedure, both DCPD and ENB microcapsules were cured at 60°C for 16 h and at 80 °C for 2 h, respectively [4]. It was observed that the microcapsules were not damaged, due to thermal tests performed at 60, 80, and 120 °C for an interval of 1, 2, 3, and 4 h [4].

Initially, by thermal treatment at 80°C, PUF-DCPD microcapsules had an average weight loss percentage of 13.93%, compared to only 9.63% in the case of MUF-ENB microcapsules [4].

Secondly, due to testing thermal stability at 120°C there was a 57.49% drop in mass for UPF-DCPD microcapsules, compared to only 19.23% for MUF-ENB microcapsules [4].

Despite the comparative analysis indicating ENB a potentially more suitable alternative, DCPD demonstrated superior compatibility with Grubbs' catalyst introduced [4]. To evaluate the healing efficiency, the microcapsules, along with the catalyst were treated at 40°C for 48 h [4]. The results showed a healing efficiency of 81% for the PUF-DCPD specimens and 77% for MUF-ENB specimens, when compared to the reference samples. Additionally, thermogravimetric analysis showed thermal degradation of 73.6% for the PUF-DCPD system and 74% for MUF-ENB [4].

Dicyclopentadiene is a colorless to yellowish liquid hydrocarbon from the pyrolysis of hydrocarbon fractions. It is insoluble in water, readily soluble in acetone, dichloromethane, ethyl acetate, n-hexane and toluene. The strongly flammable compound could be found either as a liquid, or as a crystalline solid, harmful if swallowed and toxic by inhalation, but non-corrosive.

ACGIH, The American Conference of Governmental Industrial Hygienists has attributed an eight-hour occupational exposure limit of 5 ppm, or 0.5 ppm for European exposure limits. DCPD could be transported by road, rail, and ship, being widely recommended to be stored in mild steel or stainless-steel tanks, under controlled conditions, to minimize the risk of electrostatic discharge [5].

Dicyclopentadiene is formed due to Diels-Alder addition reaction between two cyclopentadiene units, both representing a diene and a dienophile [6]. The process is reversible, the reaction allowing the spontaneous dimerization of cyclopentadiene to regenerate dicyclopentadiene in the course of several hours. At 170 °C, the monomer could be isolated by distillation of the dicyclopentadiene [6].

The dimer exists in two stereoisomers, exo- and endo-dicyclopentadiene.

Both of them exhibit chirality, a geometric property in which a molecule or ion cannot be superposed on its mirror image by any combination of rotation, translations and conformational changes.

It has been reported in the literature that exo-DCPD has higher rates in reactions like ROMP than endo-DCPD, using the Grubbs' catalyst [7]. Various issues drew attention to the low reactivity of endo-DCPD, such as unfavorable steric interactions due to its bent geometry and possible coordination between the ruthenium centre of the Grubbs' catalyst and the double bond of the dicyclopentadiene [7].

Despite the fact that, endo-isomer is the major product of the process, approximately 73%-83%, the exo-isomer of DCPD is best aligned with the experimental requirements, due to its low viscosity at temperatures reaching -50°C, high reactivity and short reaction time. Thermodynamically, the exo-isomer is approximately 0.7 kcal/mol more stable than the endo-isomer.

Catalyst

Grubbs catalyst is a ruthenium based complex used as a catalyst for olefin metathesis. Multiple Grubbs' catalyst are known in the literature, the most popular being Benzylidene-bis(tricyclohexylphosphino)-dichlororuthenium, or as its usual name Grubbs First-generation G1. In addition to this are Grubbs Second G2 and Third-generation G3 and, also, Hoveyda-Grubbs catalyst.

First-generation Grubbs catalyst was chosen, due to its various positive results in ROMP reaction. However, G2 and G3 are well-known for their higher activities compared to G1, in this particular context, G1 may react more rapidly and effectively in combination with DCPD for ROMP. Because of its smaller ligand, it contributes to a more productive reaction.

Ring-Opening Metathesis Polymerization-ROMP

Ring-Opening Metathesis Polymerization is a process where olefinic polymers are by metathesis reaction. The centre of the reaction is a Grubbs catalyst that contains a metal carbene intermediate that reacts with the olefins to form different olefinic compounds. This catalyst could split the double bond of the alkene and put the parts together with other alkenes. The ring opens and produce a different catalyst and alkene.

The mechanism contains four stages to obtain a four member metallacycle intermediate [2]. Due to the reaction of cycloalkene and a metathesis catalyst, in this case, the reaction of DCPD and Grubbs catalyst.

The second step is the opening of the metallacycle that leads to achieving a new alkene and catalyst. Then, the catalyst end reacts with a different cycloalkene producing a new metallacycle. The process continues till it forms a polymer chain [2].

ROMP reaction is facilitated by the cycloalcadiene strain. DCPD, having two available groups: the double bond of the norbornene ring and the double bond of the cyclopentene ring, could easily react and form the polymer chain, with good mechanical properties [2], [8]

4.10.3 Formulation Composition

Based on a detailed literature review, several relevant laboratory syntheses were identified. It is known that the majority of studies produced microcapsules of urea-formaldehyde, containing dicyclopentadiene, in situ polymerization technique [4], [2], [3], [9].

Materials

The most widely used materials for the situ polymerization of DCPD microcapsules include ethylene maleic anhydride copolymer(EMA) which represents the emulsifier[D1][D8], dicyclopentadiene(DCPD),the monomer. Urea and formaldehyde are used for the formation of the capsule shell in the aqueous state [4], [9] and ammonium chloride, as a hardener for the formaldehyde[D1][D8]. Sodium hydroxide has a role in rising solution pH while hydrochloric acid is lowering the solution pH [4], [9].

Resorcinol's role is to blend resin with formaldehyde and Bis(tricyclohexylphosphine)benzylidene ruthenium (IV) dichloride is the First-generation Grubbs catalyst that leads to the polymerization of DCPD [4], [9]. Additional materials would be paraffin wax and 1-octanol [9].

Matrix

The matrix used could be an epoxy(bisphenol-A), synthesized using Araldite LY556 and HY951 hardener used for the encapsulation or coating of low-

voltage components [9]. LY556 has a high-temperature resistance and exhibits efficient mechanical and dynamic properties. The Araldite LY556 and HY951 is an excellent system due to its high resistance to atmospheric degradation.

4.10.4 Coating-Substrate interface

Based on extensive review of the scientific literature, aluminum and its alloys are widely applied in a variety of industries due to their low density, high strength, high conductivity and moderate corrosion resistance [10].

Pure aluminium is highly resistant to corrosion, except in environments that present a high or low pH. It does not need a protection compared to other metals [10].

These alloys are particularly known to be used in automobile and aeronautic and space sectors. One of the most employed aluminium alloy is AA2024-T3, especially for structural application in the aeronautical industry owing to its low cost and mechanical properties.

The corrosion protection system needs to be resistant to temperatures between -55 and 80°C and stable for a long period of time [10].

Considering that they are susceptible to corrosion, the alloys were subjected to several treatments such as chemical oxidation and anodic oxidation. In addition, an efficient solution for the protection of aerospace aluminium alloys, is using organic coatings, due to their lower cost.

Protective layers such as epoxy coatings are used because of their toughness and, because it is a simple method of applying to metallic materials. The coating is commonly serves as a primer to inhibit corrosion and ensure the adhesion of the topcoat. This technique is adopted in the aerospace and military industry.

Due to literature research, a strategy for creating the chemical bond between aluminium alloy and a polymer was proposed, including two methods: Laser Engraving (LE) and Resin Pre-Coating (RPC) [11]. It was concluded that the second technique was more efficient due to the resin's low-viscosity, overcoming the capillary limitations of the adhesives [11]. LE is a high-precision method used to increase the effectiveness of the created bond [11].

The purpose of the article is to show how epoxy pins are the optimal choice for a complete adhesion to the aluminium alloy [11]. Through the structural micro-anchors the epoxy resin is blocked on the aluminium oxide and ensures a facile coating fixation [11].

For the experimental synthesis, Biphenol A epichlorohydrin was used as the epoxy resin, and JL-F30 as the laser machine. The aluminium alloys were analyzed using the laser engraving system at 24W [11].

A 10 wt% epoxy resin was mixed with acetone, the solvent used, to ensure that low viscosity. After the evaporation of the solvent, the epoxy coating was generated and successfully combined with the laser-engraved pores [11].

Several information about the experimental results are that the bond thickness was about 0.76 mm after 12 hours at room temperature, followed by a complete cross-linking after 72 hours at 60 °C was obtained [11].

The epoxy pins were successfully formed and had an average diameter of 176.2 microns [11].

Ultimately, researchers discovered that 0.3 mm is the optimal choice for the pins to be fixed [11].

The anodization is a pretreatment technique for aluminium alloys due to its simplicity. The aluminium alloy is immersed in an electrolyte with anodic polarization, resulting an anodic film composed of a thin barrier layer and an outer thick porous layer.

A typical aerospace corrosion protection system is a multi-layer system consisting of a porous anodic oxide, a corrosion inhibited organic primer and an organic coating [10].

Anodizing is an electrochemical method in which the operating conditions are extremely important to the final properties, chemistry and morphology of the anodic oxide film. The conditions include the temperature of the electrolyte and the electrode, the electrolyte pH and chemical composition and the current density [10].

One of the most important function of the anodic layer is to strengthen the adhesion of the organic coating to the metallic substrate. Adhesion implies a stable bond between oxide and the polymer [10]. In order to this adhesion functionality, it is essential to be involved physical and chemical bonds. London, Keesom and Debye interactions take place between any two molecules in contact and have the ability to generate a covalent bond or ionic bond between the oxide and the polymer [10].

Another example is acid-base interactions that form between a donor and an acceptor of electron pair. In this case the existence of hydrogen bonds is considered to be the most important factor for the adhesion between aluminium oxide and polymeric resins, particularly epoxy resins [10]. The hydroxyl groups of the epoxy resin interact with the hydroxyl groups of the aluminium oxide and generate the hydrogen bond [10].

Anodizing does not represent an independent process, being part of a complex corrosion protection system which presents pretreatments to prepare the substrate surface for the anodizing stage [10].

One first step is the elimination of any metallurgical lubricants. Cleaners and low alkaline substances are introduced to protect the aluminium [10]. The second step is alkaline etching, which removes the deformed layer near the surface formed [10]. It usually involves immersion in aqueous sodium hydroxide solution. The effect of this technique are the chemical modification and smut formation which means that insoluble elements such as iron and silicon could need further treatment to achieve a clean substrate [10].

As well as needing pretreatments in order to function, anodizing also requires post-treatments like sealing. The purpose of it is to close the open pores of the anodic oxide to ensure corrosion resistance [10].

The chemical stability, physical interfacial bonding and the strength of the interactions at the interface are fundamental factors in the high adhesion performance and corrosion stability [10].

4.10.5 Curing and Processing Parameters

The synthesis of the microcapsules has an average minimum temperature of 35°C to prevent the dicyclopentadiene monomer from crystalizing and a maximum of 50°C for the reaction of the added substances.

The curing process is estimated to have place in 24 hours at room temperature for complete solidification, but it could also be 16 hours at 60°C or 2 hours at 80°C.

The healing test would be done at 40°C for 48 hours to show the exact initial conditions.

Quality Control Methods

Several Quality Control Methods were found in literature [4], [2], [9], [10] :

FT-IR Analysis (Fourier Transform Infrared Spectroscopy)-used to verify encapsulation.

Optical Microscopy, SEM (Scanning Electron Microscopy)-confirms the spherical shape and analyze the size of microcapsules.

TEM Analysis (Transmission Electron Microscopy)- confirms the core-shell structure and measures shell thickness

Thermal Stability, TGA (Thermogravimetric Analysis)-shows mass loss as the temperature increases.

Only selected methods will be used in this study the limited access to specialized laboratories.

4.11 Sample Preparation and Handling

4.11.1 Sample Specifications

The experiment will use 9 samples made from 2017-T4 alloy of the same dimensions. In order to develop predisposition to micro-crack development, the samples will be mechanically fatigued pre-flight thus lowering the overall yielding stress by a factor. Although a proper S-N fatigue curve has not been researched for 2017-T4 alloy, an article [11] found a way of analytically determining it and the tests concluded that the method could be reliably used. For ensuring catastrophic damage will not occur, each sample will be mechanically to a maximum of 70% of the low cycle fatigue life for a particular stress. Low cycle fatigue would take place in the range of 10^4 and 10^5 cycles.

Additionally, an article [12] concluded that a brittle film on a ductile substrate will induce cracks in the substrate if the film is damaged. The article used a TiN film on a brass substrate and observed that when the film starts to crack, the ductile substrate will also form micro-cracks because of the shockwave produced by the cracking. This can help produce more micro-cracks at the surface level. This will also be implemented into our experiment preparation as the samples will have a hard anodizing operation performed. The Al_2O_3 oxide, also known as alumina, is brittle by nature and that will ensure micro-cracks have a higher chance of appearing. Furthermore, alumina has a CTE of $6.4 \mu\text{m/m}^\circ\text{C}$ while the aluminium substrate has a CTE of $23.6 \mu\text{m/m}^\circ\text{C}$. This difference will result microcracks in the alumina film before reaching the yielding stress of it. Hard anodizing the samples with a layer between 20 – 50 μm would ensure the layer would have a mechanical role not just corrosion resistance properties. Anodizing the samples will also act as an electrical isolation for the semiconductor film. The anodizing operation is detailed in chapter 4.10.4.

4.11.2 Surface Preparation Sequence

The surface preparation sequence establishes the layered structure required for the self-healing event and crack detection mechanism. The process involves three main stages: controlled oxide formation, semiconductor film deposition, and coating application.

The aluminum alloy substrate first undergoes controlled anodization to form an aluminum oxide layer. The oxide layer is significantly more rigid and more brittle than the underlying aluminum. During thermal stress, aluminum tends to expand and contract more than the oxide layer, generating internal stresses at the interface. If the thermal expansion mismatch becomes sufficiently large, micro-cracks initiate in the oxide layer. These cracks will propagate (assuming correct layer dimensions and thermal system calibration) into the polymer coating that contains the microcapsules.

A semiconductor film, several tens to hundreds of nanometers thick, will be deposited over the anodized layer. This film serves as the sensing element for crack detection. A small voltage will be applied across the semiconductor layer and resistance will be measured periodically. The semiconductor film exhibits stable initial electrical resistance and when a microcrack initiates in the

anodized layer, it propagates through the semiconductor film toward the coating. If polymerization occurs in the coating, partial crack closure may result in measurable resistance changes. Preliminary trials for semiconductor deposition have begun at the INFLPR plasma laboratory; film thickness and deposition parameters remain under evaluation. The self-healing coating is applied over the semiconductor layer.

4.11.3 Pre-flight Handling and Shipping

Samples are stored in a controlled environment at room temperature (15-25 °C). To avoid premature unintentional microcracking before the flight, the storage facility isolates the samples from mechanical shocks and high-frequency vibrations.

The experiment will be sent to Esrange Space Center packed in a Pelican Protector 1620 case. To ensure thermal stability, we use Vacuum Insulated Panels (VIPs) between the case walls, and a closed-cell PE foam insert with hard-shell 22°C PCM bricks to maintain a constant internal temperature. Sorbothane dampening pads envelope the experiment box, absorbing omnidirectional impacts and vibrations during transit. To protect against moisture and static, the equipment will be sealed in an electrostatic discharge (ESD) moisture barrier bag with silica gel packs.

This setup is designed to shield the experiment from environmental conditions, vibration, and impact during transport and is kept at a 15-25°C climate.

A specialized international freight courier (e.g., FedEx, TNT, DHL Express Dangerous Goods) will oversee the shipment. The outer package will be clearly marked with "SCIENTIFIC SAMPLES - FRAGILE" labels and will include all necessary Safety Data Sheets (SDS) for customs clearance.

The coating will be applied to the samples in Romania at The Petru Poni Macromolecular Institute. Since the specimens will be flight-ready upon arrival, no chemicals will be handled or applied at the Esrange facilities.

4.12 In-Flight Crack Generation and Observation System

4.12.1 Experimental Matrix

The experimental samples are organized in three groups of three plates each. The matrix investigates the effect of thermal stress and mechanical preconditioning on microcrack formation and subsequent self-healing behavior.

Group	No. of plates	Temp.	Fatigue	Monitored plate
1	3	T1	baseline	1
2	3	T2	baseline	1
3	3	T2	high	1

Temperature setpoints T1 and T2 are determined from ground characterization of fatigue-pretreated samples. T1 corresponds to the thermal threshold for crack initiation; T2 produces robust microcracking without risk of substrate yielding or coating delamination. Group 3 uses the T2 temperature with increased fatigue cycle preconditioning to isolate the effect of mechanical damage density on healing performance.

One plate per group includes electrical monitoring due to power budget constraints. The six unmonitored samples provide post-flight statistical validation via scanning electron microscopy.

At the time of writing, preliminary thermal cycling tests on fatigue-pretreated samples are pending. Temperature values T1 and T2 will be established following completion of these tests and documented in the future revision of this document.

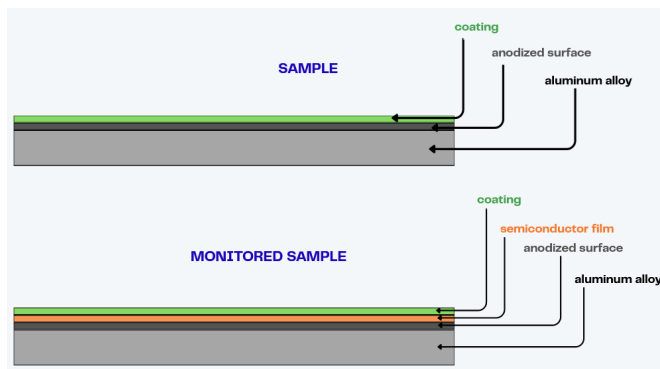


Figure 4.12.1 - Coating layers

4.12.2 Thermal Crack Generation Mechanism

As detailed in the Mechanical Design and 4.11 chapters, the crack generation mechanism exploits thermal expansion mismatch between the aluminium substrate, anodized layer, and polymer coating. Controlled heating of the cold specimens induces tensile stress at the coating-substrate interface, promoting microcrack nucleation and propagation through the coating thickness.

The heating system uses zonal control per specimen group, with three independent zones each managing three specimens. Heating is achieved by polyamide flexible heaters bonded to the underside of the aluminium baseplate, with PID temperatures control.

At the moment of writing, preliminary experimental validation of the crack generation mechanism will be conducted using flight-representative specimens. Due to manufacturing lead times for the test samples, these tests will be completed until the Action Matrix v1 deadline.

The crack generation system is intended to be passive and autonomous, requiring no external intervention once activated by the flight software.

4.12.3 Electrical Observation

The electrical observation system monitors crack formation and propagation through continuous resistance measurements across the semiconductor film. Due to power budget constraints, only three of nine samples include this detection capability, one witness sample from each thermal group.

A low-current, low-voltage signal will be applied across the semiconductor film. Crack propagation interrupts the electrical path, producing measurable resistance changes that indicate crack initiation and propagation. Multiple thin semiconductor strips can be deposited and connected to separate resistance measurements channels to improve crack localization data.

Each monitored sample connects to two terminals for resistance measurement. The data acquisition system records resistance values with timestamps for correlation with thermal cycling data and ground-based analysis. The software stores resistance readings at defined intervals alongside temperature data.

INFLPR has agreed to deposit the nanometric semiconductor film over the anodized surface. Preliminary deposition trials are underway with the plasma deposition machine to evaluate layer thickness, deposition time, and surface roughness parameters. Silver (Ag) was selected as the semiconductor material as it is already available at the institute.

5 EXPERIMENT VERIFICATION AND TESTING

5.1 Verification Matrix

Table 5-1: Verification table

ID	Requirement text	Method	Reference	Status	Verification Result
F.1	The experiment shall accommodate nine coating specimens (50 x 30 x 3 mm).	I		To be done	
F.2	The experiment shall induce controlled microcrack damage in coating layers through thermally induced stress.	T	T-01	To be done	
F.3	The experiment shall generate microcracks with characteristic dimensions compatible with self-healing microcapsule activation (width 10-50 μm , depth 50-150 μm).	I		To be done	
F.4	The experiment shall measure ambient pressure throughout the flight profile.	T	T-02	To be done	
F.5	The experiment shall record ambient pressure measurements with timestamps throughout the flight profile.	T	T-02	To be done	
F.6	The experiment shall measure ambient temperature throughout the flight profile.	T	T-03	To be done	
F.7	The experiment shall record ambient temperature measurements with timestamps throughout the flight profile.	T	T-03	To be done	

F.8	The experiment shall measure ambient ultraviolet radiation intensity throughout the flight profile.	T	T-04	To be done	
F.9	The experiment shall record ambient ultraviolet radiation measurements with timestamps throughout the flight profile.	T	T-04	To be done	
F.10	The experiment shall acquire temperature data from each individual coating specimen.	T	T-05	To be done	
F.11	The experiment shall store all acquired data to non-volatile memory onboard.	T	T-05 T-06	To be done	
F.12	The experiment shall transmit housekeeping data via the BEXUS E-Link interface during the flight.	T	T-07	To be done	
F.13	The experiment shall transmit housekeeping data at minimum 1 Hz update rate.	T	T-07	To be done	
F.14	The experiment shall control the heater system to achieve predefined temperature profiles required for microcrack generation.	T	T-05	To be done	
F.15	The heater system shall achieve a target temperature of 70 °C.	T	T-05	To be done	
F.16	The experiment shall support an optional secondary heating cycle if flight duration permits specimen cooldown.	T	T-05	To be done	

F.17	The heater system shall implement fail-safe shutdown if temperature exceeds 100 °C.	T	T-08	To be done	
P.1	Ambient temperature shall be measured over range -90°C to +50°C.	R		To be done	
P.2	Ambient temperature shall be measured with accuracy of ±3°C.	R		To be done	
P.3	Ambient pressure shall be measured over range 5 mbar to 1050 mbar.	R		To be done	
P.4	Ambient pressure shall be measured with accuracy of ±1 mbar.	R		To be done	
P.5	Ambient ultraviolet radiation shall be measured over wavelength range of 350 nm to 820 nm.	R		To be done	
P.6	Ambient ultraviolet radiation shall be measured with accuracy of ±5%.	R		To be done	
P.7	Coating specimen temperature shall be measured with accuracy of ±2°C.	R		To be done	
P.8	Heater system shall achieve a temperature ramp rate of 5 °C/min minimum.	R, T	T-05	To be done	
P.9	Heater system shall maintain specimen temperature uniformity within ±2°C across all specimens.	R		To be done	
P.10	Heater system shall sustain target temperature for 10 minutes minimum.	R, T	T-05	To be done	

D.1	The experiment shall comply with the mechanical interface requirements of the BEXUS gondola.	R, I		To be done	
D.2	The experiment shall comply with the electrical interface requirements of the BEXUS gondola.	R, I		To be done	
D.3	The experiment shall comply with the BEXUS E-Link communication interface.	R, T	T-07	To be done	
D.4	The experiment shall withstand the vibration environment specified for the BEXUS launch and flight.	T	T-09	To be done	
D.5	The experiment shall withstand the pressure environment of the BEXUS flight profile.	A, T	T-10	To be done	
D.6	The experiment shall withstand the thermal environment of the BEXUS flight profile.	A, T	T-11	To be done	
D.7	The experiment shall not disturb or harm the launch vehicle or other experiments.	R		To be done	
D.8	Components introduced into thermal-vacuum chamber shall be verified to be compatible with the facility requirements of the National Institute of Space Sciences.	T, R, I	T-10	To be done	
D.9	All components containing stored energy shall be qualified for use on a BEXUS balloon.	R		To be done	
D.10	The experiment shall accommodate 9 specimens with dimensions of 50 mm x 30 mm x 3 mm.	R, I		To be done	

D.11	The experiment shall withstand a horizontal load of 5G.	A		To be done	
D.12	The experiment shall withstand a vertical load of 10G.	A, T	T-12	To be done	
O.1	The experiment shall be activated prior to launch via ground command or predefined autonomous sequence.	T	T-17	To be done	
O.2	The experiment shall execute heating sequence for microcrack generation autonomously during flight.	T	T-10	To be done	
O.3	The experiment shall initiate heating when ambient pressure falls below 100 mbar.	T	T-10	To be done	
O.4	The experiment shall continue data acquisition autonomously in case of loss of ground communication.	T	T-12	To be done	
O.5	The experiment shall support multiple operational modes: STANDBY, RUN, SAFE.	T	T-13	To be done	
O.6	The experiment shall retain all stored data after landing without external power for a minimum of 48 hours.	T	T-14	To be done	
O.7	The experiment shall allow access to the coating specimens during ground operations without disassembly of the gondola mounting interface.	T	T-15	To be done	

O.8	The experiment shall accept a request for radio silence at any time while on the launch pad.	T	T-16	To be done	
O.9	The experiment shall provide a visual status indication during ground operations.	T	T-17	To be done	
O.10	The experiment shall log heating cycle data including start time, peak temperature, hold duration, and cooldown rate.	T	T-05	To be done	
O.11	The experiment downlink data rate shall not exceed 100 kbps.	T	T-18	To be done	

5.2 Verification Plan

Table 5-2: T-01 – Terahertz NDT Test

Test number	T-01
Test type	NDT
Test facility	INFLPR
Tested item	Samples with applied coating
Model	Prototype
Procedure, Test level and duration	Coating thickness will be measured with submicron accuracy, and subsurface defects will be detected using terahertz sensors. This NDT uses electromagnetic radiation from 0.1 to 10 THz to inspect, image, and analyze materials damage and chemical compositions.
Test campaign duration	1 day
Test campaign date	June 2026
Test completed	NO

Requirements verified	F.2 - NO
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Table 5-3: T-02 – Pressure sensor testing

Test number	T-02
Test type	Functional
Test facility	UPB
Tested item	Pressure sensor
Model	Prototype
Procedure, Test level and duration	Proceed to measure specific ambient pressure conditions with external reference sensors to verify the sensor's functionality with whole setup. Track accuracy and measurement rate of ambient pressure in data logs.
Test campaign duration	1 day
Test campaign date	June 2026
Test completed	NO
Requirements verified	F.4 - NO F.5 - NO

Table 5-4: T-03 – Temperature sensor and compatibility with flight conditions test

Test number	T-03
Test type	Functional
Test facility	UPB
Tested item	Whole experiment
Model	Prototype
Procedure, Test level and duration	Acceptance-level thermal test consists of exposure to temperatures between –40°C and +60°C, including three thermal cycles with a minimum soak time of 30 minutes at each extreme Proceed to measure specific ambient temperature with external reference sensors to verify the sensor's functionality while whole setup is integrated
Test campaign duration	1 day

Test campaign date	June 2026
Test completed	NO
Requirements verified	F.4 - NO F.6 – NO F.7 – NO D.6 - NO

Table 5-5: T-04 – UV sensor characterization test

Test number	T-04
Test type	Functional
Test facility	UPB
Tested item	Ultraviolet radiation sensing system
Model	Flight
Procedure, Test level and duration	Expose the UV sensor to a low intensity UV lamp and verify whether the sensors measure ultraviolet radiation. Perform acquisition for 1 hour, stop the test and after verify if the UV measurement data was recorded on the onboard memory.
Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO
Requirements verified	F.8 – NO F.9 - NO

Table 5-6: T-05 – Individual specimen heating and temperature data acquisition and manipulation

Test number	T-05
Test type	Functional
Test facility	UPB
Tested item	Heating system
Model	Prototype
Procedure, Test level and duration	Record the initial temperature of all specimens at ambient conditions; Use the heating system to apply a controlled temperature increase up to 70 °C; Monitor temperature readings from each specimen

	during heating, calculate and record temperature ramp rate; Keep samples at a sustained temperature for 10 min; Selectively heat or insulate one specimen in order to confirm individual channel response. After specimen cooldown, procedure is to be repeated.
Test campaign duration	2 days
Test campaign date	June 2026
Test completed	NO
Requirements verified	F.11 – NO F.14 – NO F.15 – NO F.16 – NO P.8 – NO P.10 – NO O.10 – NO

Table 5-7: T-06 – Data storage acquisition test

Test number	T-06
Test type	Functional
Test facility	UPB
Tested item	Non-volatile memory device
Model	Prototype
Procedure, Test level and duration	Operational sequence test verifying data acquisition: Proceed memory initialization (memory should report empty or initialized state); start data acquisition by command and generate a known data pattern and allow acquisition to run for 1000 samples (acquisition status shall be nominal and sample counter shall increase monotonically); stop data acquisition (acquisition shall stop without error and data storage process shall be completed); after power interruption and data readback, retrieved data shall match exactly the original injected pattern.
Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO

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Field Code Changed

Requirements verified	F.11 - NO
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Table 5-8: T-07 – E-Link communication and telemetry test

Test number	T-07
Test type	Functional
Test facility	UPB
Tested item	Communication and telemetry subsystem
Model	Prototype
Procedure, Test level and duration	Interface test verifying correct and timely transmission of housekeeping data via the communication interface, including compliance with update data requirements and radio silence command handling
Test campaign duration	1 day
Test campaign date	June 2026
Test completed	NO
Requirements verified	F.12 – NO F.13 – NO D.3 – NO

Table 5-9: T-08 – Fail-safe shutdown implementation test

Test number	T-08
Test type	Software
Test facility	UPB
Tested item	Overheating prevention logic
Model	Prototype
Procedure, Test level and duration	Use the heating system to apply a controlled temperature increase up to 70 °C; Apply an external thermal load, in order to increase temperature over 100 °C; When sensing temperatures over 100 °C, the experiment shall automatically shut down the heaters to protect the experiment.
Test campaign duration	2 hours

Test campaign date	June 2026
Test completed	NO
Requirements verified	F.17

Table 5-10: T-09 – Vibration test

Test number	T-09
Test type	Vibration
Test facility	National Institute of Space Sciences
Tested item	Whole experiment
Model	Prototype
Procedure, Test level and duration	Acceptance-level vibration test simulating vibration values in the specified range listed in the BEXUS Manual
Test campaign duration	2 hours
Test campaign date	August 2026
Test completed	NO
Requirements verified	D.4 - NO

Table 5-11: T-10 – Pressure / vacuum test

Test number	T-10
Test type	Vacuum / pressure
Test facility	National Institute of Space Sciences
Tested item	Whole experiment
Model	Prototype
Procedure, Test level and duration	Acceptance-level pressure test reproducing the BEXUS ascent and float pressure profile within a controlled vacuum chamber; Due to pressure loss, under 100 mbar, the heating sequence shall start autonomously. Total mass loss due to outgassing shall be verified.
Test campaign duration	1 day
Test campaign date	June 2026
Test completed	NO

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Field Code Changed

Requirements verified	D.5 – NO D.8 – NO O.2 – NO O.3 – NO
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Table 5-12: T-11 – Static Vertical Load Test

Test number	T-12
Test type	Structural Load Verification
Test facility	UPB
Tested item	Whole experiment
Model	Prototype
Procedure, Test level and duration	Test flight model by applying static load equal to 10 times payload weight for 5 minutes; Distribute load via gondola interface brackets.
Test campaign duration	1 day
Test campaign date	August 2026
Test completed	NO
Requirements verified	D.12 – NO

Table 5-13: T-12 – Autonomous Data Acquisition During Loss of Signal Test

Test number	T-12
Test type	Software
Test facility	UPB
Tested item	Software, Storage Memory, and Communication Subsystem
Model	Prototype
Procedure, Test level and duration	Start normal data collection while connected to the Ground Station using E-Link. Next, unplug the Ethernet cable for 15 minutes to simulate a complete signal loss. After 15 minutes, plug the cable back in, download the SD or USB logs, and check that the sensor data was recorded accurately and without any gaps during the blackout.

Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.4 – NO

Table 5-14: T-13 – Operational Mode Transitions

Test number	T-13
Test type	Software
Test facility	UPB
Tested item	Software and Communication Subsystem
Model	Prototype
Procedure, Test level and duration	Send sequential telecommands from the Ground Station to transition the active payload through STANDBY, RUN, and SAFE modes. Verify via telemetry flags and hardware power draw that the software successfully enters each state and executes the correct operational logic
Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.5 – NO

Table 5-15: T-14 – Data Retention when Unpowered Test

Test number	T-14
Test type	Functional
Test facility	UPB
Tested item	microSD card
Model	Prototype
Procedure, Test level and duration	First, save a known set of test data to the onboard microSD card. Next, disconnect all batteries and external power sources from the payload for at least 48 hours. After 48 hours, check the SD card to make

	sure all the data is still there and has not been corrupted or lost.
Test campaign duration	3 days
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.6 – NO

Table 5-16: T-15 – Ease-of-access to specimens test

Test number	T-15
Test type	Functional
Test facility	UPB
Tested item	Coating Specimens
Model	Flight
Procedure, Test level and duration	Mount the flight model to the gondola interface and attempt to access the coating specimen compartment following the ground operation procedure. Confirm that the specimens can be removed, replaced, or adjusted if required by the ground handling procedure.
Test campaign duration	1 hour
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.7 – NO

Table 5-17: T-16 – Radio Silence Compliance Test

Test number	T-16
Test type	Functional
Test facility	ESRANGE (before flight)
Tested item	Whole experiment
Model	Flight
Procedure, Test level and duration	Establish communication between the experiment and the ground control system via the BEXUS

	communication interface. Confirm that telemetry data is transmitted. Issue a radio silence request command from the ground control system. Confirm that the experiment remains operational internally while communication transmissions are disabled.
Test campaign duration	1 hour
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.8 – NO

Table 5-18: T-17 – Visual status indicator test

Test number	T-06
Test type	Functional
Test facility	ESRANGE (before flight)
Tested item	Whole experiment
Model	Prototype
Procedure, Test level and duration	Power on the experiment in nominal ground operation mode. Observe the visual status indicator immediately after startup. Command the experiment to enter different operational modes. Observe the visual indicator during each operational mode and verify that the visual indicator changes according to the corresponding experiment status. Confirm that the indicator remains visible and readable from a distance of 10 meters.
Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.9 - NO

Table 5-19: T-18 – Downlink Data Rate Test

Test number	T-18
Test type	Software
Test facility	UPB

Tested item	Payload structural frame
Model	Prototype
Procedure, Test level and duration	Run the experiment in full operational mode with all sensors streaming and monitor the E-Link interface using a network analyzer (e.g., Wireshark). Confirm that the total outgoing traffic, including packet headers and protocol overhead, consistently remains below the 100 kbps limit.
Test campaign duration	2 hours
Test campaign date	June 2026
Test completed	NO
Requirements verified	O.11- NO

5.3 Verification Results

Table ... represents the standard template that will be used after the execution of every verification activity. For each test it records the verification number, the type of test, the verified item, the summarized procedure, and the quantitative results. If a non-conformance occurs, the completed sheet will serve as the initiating document for any review meeting, ensuring that any anomaly will be tracked with closure.

The following format will be used for all qualifications and sub system test, and this section will be updated with formal results.

Table ...: – Test Report Template

Verification number	T-16
Test type	Structural Load Verification
Test facility	UPB
Tested item	COATHEAL Payload structure, gondola mounting interface with weight
Verification description	Apply 10 times payload mass for 5 minutes, confirm that no major size change occurred after
Expected results	<±2.5 mm displacement is tolerated, no fastener loosening and no deformation are accepted

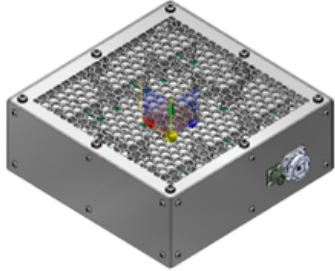
Obtained results	Payload was able to sustain 15 minutes of time tested under the specified test conditions (50 kg of weight) and had no conformities or deformation greater than 1mm.
Conclusions	Payload, frame, and mounting hardware functionality and structure are not damaged and are qualified for flight.

6 LAUNCH CAMPAIGN PREPARATION

6.1 Input for the Campaign / Flight Requirement Plans

6.1.1 Dimensions and Mass

Table 6-1: Experiment Mass and Volume

Experiment mass (in kg):	Total weight: 5,071kg With a safety margin: 5,816kg
Experiment dimensions (for BEXUS) (in m):	w x h x L (0.258 x 0.100 x 0.268) m Including aluminium profiles and mounting brackets: (0.600 x 0.139 x 0.308) m
Experiment expected COG (centre of gravity) position:	Approximately in the center of the box 

6.1.2 Ground and Flight Safety Risks

Table 6-2: Ground and flight safety risks

Risk	Key Characteristics	Mitigation
Incorrect assembly	Incorrect assembly of the experiment.	Following strict assembly procedures.
Bolt loosening	Bolt loosening due to vibrations.	Precise bolt pre-load calculation. Using torque wrench for tightening.

Structural failure of the aluminium samples	Structural failure of the aluminium samples due to fatigue cycling and incorrect handling.	Fatigue cycling with respect to a predetermined safety factor. Ensuring careful handling.
COATHEAL samples contamination	COATHEAL samples contamination during transport to Esrange and pre-flight integration.	Ensuring good logistic conditions. Checklist for standard operating procedures for COATHEAL application.

6.1.3 Electrical Interfaces

Table 6-3: Electrical interfaces applicable to BEXUS

BEXUS Electrical Interfaces		
E-Link Interface: E-Link required? Yes/No		
	Number of E-Link interfaces:	1
	Number of required IP addresses:	2
	Data rate – downlink (max. and average):	100 Kbit/s - 1.2 Kbit/s
	Data rate – uplink (max. and average):	100 Kbit/s - 0.2 Kbit/s
	Interface type (RS-232, Ethernet):	Ethernet
Power system: Gondola power required? Yes		
	Peak power and current consumption:	48.23 W and 1.72 A
	Average power and current consumption:	27.92 W and 0.82 A
	Total power and current consumption after lift-off	223.9 Wh and 4.833 Ah
Power system: Experiment includes batteries? No		
	Type of batteries:	
	Number of batteries (and S x P):	
	Capacity (1 battery):	Ah
	Voltage (1 battery):	V

6.1.4 Launch Site Requirements

The experiment is expected to require standard integration facilities, including basic workspaces, power access, and common laboratory tools. In addition to standard hand tools, the team is currently considering the use of cordless screwdrivers to support assembly and disassembly activities during the campaign. Protective transport and storage solutions, such as Pelican-type cases, are preliminary foreseen for the handling and storage of samples before and after flight. Furthermore, the use of a microscope is under consideration to enable rapid post-flight visual inspection of the coating specimens directly at the launch site.

The exact list of required equipment tools, and consumables, as well as quantities, will be finalized and communicated at later project stages once the experiment configuration and operational procedures are fully defined.

6.1.5 Flight Requirements

Table 6-4: Flight Requirements applicable to BEXUS

Optimal altitude	≥ 15 km
Preferred path	Nominal BEXUS flight profile: ascent, float, descent
Minimum float time	≥ 1 hour
Ground track length	No specific requirement
Light/dark condition	No preference*

*After consultations with RISE experts and former SOBER and SOLDERx members, we concluded that temperatures are not as low as expected thus making the sample frame hard to keep below 0°C during daytime. Therefore, we consider flying during the night time.

6.1.6 Accommodation Requirements

The experiment requires a spot on the gondola where the box can sit flat with its protective mesh (top) lid facing the sky, with access to the external environment.

6.2 Preparation and Test Activities at Esrange

An initial overview of the operational flow during the launch campaign is provided as following:

The main activities during the launch campaign include:

- testing communication between the OBC and Ground Station
- testing communication between the OBC and sensors
- position the experiment samples in the dedicated slots
- mount the experiment box

At the start of each day, a team meeting is expected, during which the planned tasks should be outlined. At the end of the day, team debriefing is foreseen, during which the activities performed will be summarized and any issues,

anomalies, or unexpected observations encountered during the day will be discussed within the team.

All team members participating in the launch campaign will comply with Esrange and EuroLaunch safety regulations at all times.

The preparation and test activities are planned to be conducted following preliminary checklists prepared before arrival at Esrange, which will be written as the experiment progresses.

6.3 Timeline for Countdown and Flight

At the current stage, the detailed timeline for countdown and flight operations is still to be defined. The experiment is designed to operate in a fully autonomous mode during flight.

The experiment is expected to be powered prior to launch and to execute its internal operational sequence autonomously during ascent, float, and descent phases. No in-flight commanding from the ground is currently planned.

A detailed and time-resolved timeline for countdown and flight operations will be provided at a later project stage.

6.4 Post-Flight Activities

6.4.1 Recovery Operations

Recovery instructions

1. Unplug the electrical connector and Ethernet connector from the side
2. Using the 3mm hex driver provided, remove the four M5 (ISO 7380-1) screws located on the bottom of the experiment box. These screws attach the box to the strut profile bars.

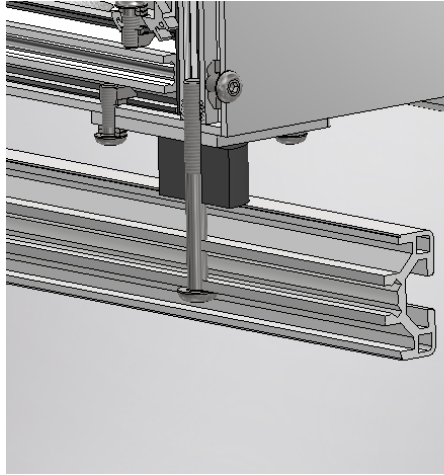


Figure 6.4.1 - M5 bolts in strut profiles

3. Place the experiment box into the transport case we provide. This case will protect the equipment during shipping.

6.4.2 Initial Inspection Procedures

Following recovery, a non-invasive visual inspection of the samples will be conducted using a portable digital microscope (Dino Lite AM7115MZT). The inspection will identify macro cracking, delamination, spiderweb-like fracture patterns, localized chipping, and deep stress lines. All observations will be recorded on-site. The final evaluation of the experiment will rely on Scanning Electron Microscopy analysis. To preserve sample condition, the specimens will not undergo additional handling and will be returned directly to the laboratory in Romania.

6.4.3 Return Transport Preparation

We will return the experiment box in the original Pelican Protector 1620 case utilized for its initial shipment.

An international freight service authorized to transport Class 9 dangerous goods, such as FedEx, TNT, or DHL Express Dangerous Goods, will ship the box from Esrange Space Center in Kiruna to the laboratory in Romania. The container will be clearly labeled "SCIENTIFIC SAMPLES - FRAGILE" and will include all required safety data sheets (SDS) to facilitate customs clearance.

Upon dispatch, the logistics provider will deliver the shipment directly to the designated facility. The case will remain sealed and will not be opened or inspected during transit

Case with experiment inside characteristics:

- 110 -

Dimensions	63x49.2x35.2 cm
Weight	17kg

7 DATA ANALYSIS AND RESULTS

7.1 Data Analysis Plan

After data extraction, preprocessing, and processing, the observed environmental parameters (temperature, pressure, and altitude) will be correlated with optical response signals obtained from the single UV photodiode. This correlation of multiple measurements will enable the identification of damage initiation, microcrack propagation, and potential healing onset during specific phases of the flight. The collected data will be entered into a table and organized into categories.

In the laboratory, multiple tests will be performed such as SEM (Scanning Electron Microscopy) imaging for examining microstructural healing and crack closure, XPS (X-Ray Photoelectron Spectroscopy) for identifying polymer fill and confirming catalyst signature and optical microscopy and profilometry for crack measurement and thickness change.

X-Ray Photoelectron Spectroscopy will start after samples are handled with gloves to prevent contamination and no conductive coating is applied, surface is cleaned by bombardment with inert gas ions to remove loose particles, then samples are mounted on conductive holders using carbon tape and electrical grounding is ensured to minimize charging. Also, before analysis, ultra-high vacuum conditions are achieved. Initially, a wide energy scan (0-1200 eV) is done to identify all surface elements and to confirm the presence of expected elements. Secondly, high-resolution scanning focuses on C (Carbon) 1s (polymer backbone and functional groups), O (Oxygen) 1s (oxidation, crosslinking, healing reactions), and catalyst-specific peaks. Other parameters analyzed by XPS are confirmation of polymer infiltration into cracks, identification of catalyst residues or active species, chemical bonding states, and evidence of curing reactions.

Scanning Electron Microscopy will be performed after the samples are cut to expose the region containing the crack, cleaned with ethanol or isopropanol to remove surface contaminants, dried under nitrogen flow to avoid residue deposition and coated with a thin conductive layer (5-10 nm of gold, platinum or carbon using sputter coating). During the initial scan, low magnification imaging to locate cracks and healed regions and identification of reference points for before and after healing comparison are being done. Lastly, high magnification imaging (with magnifications from 1000x to 50000x) is used to observe crack edges, identify polymer flow into cracks, detect bridging, filling or sealing of cracks.

Optical microscopy is performed right after sample preparation, which includes light polishing to enhance contrast at crack edges and cleaning of surface dust using compressed air or ethanol wipe. The magnification range used for analysis of the surface is 50x-500x. Cracks are identified and imaged before healing (reference images are stored for quantitative comparison). Same regions are reimaged after healing treatment, and alignment is ensured using surface markers or sample edges. Observed parameters through this method

are crack length and width, degree of visible crack closure, uniformity of healing along crack path and presence of residual voids or discoloration.

Profilometry (surface profiling) is made after samples are mounted securely to avoid tilting, and an undamaged area is selected as a baseline reference. Through profilometry, a pre-healing scan is performed (crack depth and profile are recorded) and a post-healing scan (same scan path is repeated, and direct comparison is accomplished) is achieved. Moreover, crack depth reduction and thickness increase due to polymer fill and healing uniformity across scanned regions are analyzed by using the profilometry method.

7.2 Launch Campaign

The flight preparations can be found in Chapter 6.

The launch campaign constitutes the final phase of experiment verification and qualification, encompassing mechanical and electrical integration, functional testing, and operational validation prior to flight at Esrange Space Center.

Prior to launch, the experiment will undergo comprehensive on-site functional testing. These activities include power-on verification, validation of the data acquisition system, confirmation of sensor operability, and verification of onboard data storage redundancy.

The resistive heating elements integrated into the aluminum panels will be tested during the campaign to confirm controlled thermal cycling capability under imposed power conditions. Thermal readiness of the electronics enclosure will be assessed by combined passive insulation inspection and active heater testing, ensuring that all components remain within their qualified operating temperature limits during the expected flight phases.

Mechanical integration will include verification of structural integrity, fastening security, and vibration robustness to ensure survivability during ascent, float, and descent phases. Electrical integration will be performed using the standard BEXUS 28 V DC power supply and the E-link communication interface.

The autonomous operational sequence, including startup logic, data logging, telemetry transmission, and safe shutdown behavior, will be verified prior to final gondola integration.

The launch campaign will conclude with a post-flight assessment, documenting experiment performance, deviations from expected behavior, and operational lessons learned relevant to future missions.

7.3 Results

Based on results reported in the scientific literature for microcapsule-based self-healing coatings exposed to low temperature and reduced pressure

environments, the COATHEAL experiment is expected to demonstrate a detectable healing response following induced damage. When the experiment reaches the stratosphere, during thermal cycling, mechanical cracking initiates capsule rupture and monomer release. At the same time, local temperature and catalyst activation are enhanced by UV exposure, which leads to faster ROMP curing and sealing of the damaged area.

Post-flight analysis will focus on the qualitative assessment of the coating condition using optical microscopy and SEM. Indicators of successful healing include partial crack closure and evidence of polymerized material within the damaged region.

The experiment aims to demonstrate the occurrence and feasibility of the self-healing mechanism under flight-relevant conditions. Successful performance of the experiment will be defined by the observation of polymerized material bridging damaged regions and by the structural integrity of the coating. These success criteria will be defined as we advance in programme phases and will be assessed through comparative pre-flight and post-flight inspection of the samples.

7.4 Input to Campaign Report

To be completed.

7.5 Lessons Learned

7.5.1 Experiences and challenges

During the project

- the team experienced the beginning of the learning curve related to space project standards and procedures, particularly the ECSS framework
- differences in prior experience among team members required additional effort to align expectations, working methods and technical language
- challenge of recovering progress lost due to a former teammate

During early experiment development

- early phases showed the importance of clear task definition and ownership, especially when multiple subsystems are developed in parallel

- the preparation of technical documentation revealed that design maturity must progress in parallel with documentation, not sequentially
- limited prior exposure to space-oriented constraints (materials, safety, mass, volume) required a shift from conventional engineering assumptions toward near-space engineering thinking

7.5.2 Identified failures and mistakes

- Insufficient early communication of availability and workload, which reduced predictability in planning and resource allocation
- Initial tendency to approach the project with terrestrial engineering assumptions, particularly regarding materials, thermal behavior and environmental constraints / conditions (such as low pressure)
- limited early cross-review between subsystems, which delayed identification of dependencies and interfaces
- underestimating the effort required for programme-compliant documentation at early stages, leading to time pressure during milestone preparation
- collective mistake of leaving tasks until the last minute for various reasons, resulting in AI use to accelerate writing
- mistake of having one concept in mind but documenting another or not writing with sufficient detail

7.5.3 Possible improvements

Improvements to the experiment development

- maintain a continuously updated and shared documentation repository to ensure traceability and consistency
- introduce more internal design reviews

Improvements to internal team management

- establish clearer communication protocols
- ensure knowledge transfer across the team earlier
- encourage more early discussions of uncertainties and open questions to avoid late-stage bottlenecks
- communication has improved
- several team members were removed due to inactivity; the collective learned from past coordination errors and now collaborates more closely, including regular calls during experiment work and joint visits to institutes

7.5.4 Skills and knowledge gained

- team members gained introductory experience with ECSS standards and the structure of the Programme
- several members acquired beginner-level proficiency in ANSYS Thermal
- increased understanding of polymer science and coating chemistry, including material selection for near-space conditions
- enhanced awareness of the differences between the “everyday” terrestrial engineering and space/near-space engineering, especially regarding materials, environmental loads, and safety requirements
- improved understanding of the roles of programme organizers
- several team members learned more about CNC machining, specifically how design must account for milling head capabilities
- several team members gained knowledge of polymer chemistry and materials through literature study, institute visits, and consultations with professors

8 ABBREVIATIONS AND REFERENCES

8.1 Abbreviations

3D	3-Dimensional
ACGIH	The American Conference of Governmental Industrial Hygienists
ADC	Analog Digital Convertor
BA	Bachelor of Arts
BEng	Bachelor of Engineering
BSc	Bachelor of Science
CAD	Computer Aided Design
CDR	Critical Design Review
CERC	International Students Conference
CNC	Computer Numerical Control
COATHEAL	In-Flight Demonstration of Self-Healing Coatings
COG	Centre of Gravity
COTS	Commercial-Off-The-Shelf
CRP	Campaign Requirement Plan
CTE	Coefficient of Thermal Expansion
DC	Direct Current
DCPD	Dicyclopentadiene
DLR	Deutsches Zentrum für Luft- und Raumfahrt
DVI	Detailed Visual Inspection
EAR	Experiment Acceptance Review
ECSS	European Cooperation Space Standardization
ENB	5-ethylidene-2-norbornene
ESA	European Space Agency
ESC	Echo School of Arts and Technology
ESD	Electrostatic Discharge
Espace	Espace Space Center
FEA	Finite Element Analysis
FEM	Finite Element Method
FR4	Flame Retardant 4 (garolite)
G1	Grubbs First Generation
G2	Grubbs Second Generation
G3	Grubbs Third Generation
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit
ICD	Interface Control Document
ICOS	Institute of Supramolecular and Organic Chemistry
INCAS	National Institute of Research in Aerospace Sciences
INFLPR	National Institute of Lasers, Plasma and Radiation Physics

IPR	Integration Progress Review
ISO	International Standardization Organization
MATLAB	Matrix Laboratory
Mbps	Mega Bits per second
MEng	Master of Engineering
MORABA	Mobile Raketen Basis (DLR, EuroLaunch)
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MoU	Memorandum of Understanding
MUF	Melamine-urea-formaldehyde
MSC	Mass Storage Class
MTA	Military Technical Academy "Ferdinand I"
NASA	National Aeronautics and Space Administration
NDT	Non-Destructive Testing
NMEA	National Marine Electronics Association
OBC	On-Board Computer
PCB	Printed Circuit Board
PCM	Phase Change Material
PDR	Preliminary Design Review
PID	Proportional Integral Derivative
POSS	Polyhedral oligomeric silsesquioxanes
PU	Polyurethane
PE	Polyethylene
PWM	Pulse Width Modulation
QR	Quick Response
RF	Radio Frequency
RISE	Romanian In-Space Engineering
RTD	Resistance Temperature Detector
SED	Student Experiment Documentation
SEM	Scanning Electron Microscopy
SD	Secure Digital
SDS	Safety Data Sheets (SDS)
SNSA	Swedish National Space Agency
SNSPA	National School of Political and Administrative Studies
SOE	Start Of Experiment
SPI	Serial Peripheral Interface
SSC	Swedish Space Corporation
SSH	Secure Shell
STEM	Science, Technology, Engineering, Mathematics
STW	Student Training Week
TBD	To be determined/decided
TCP	Transmission Control Protocol
TCS	Temperature Control System
TS	Temperature Sensor

UART	Universal Asynchronous Receiver/Transmitter
UPB	University Politehnica of Bucharest
USB	Universal Serial Bus
UNIBUC	University of Bucharest
UVS	UltraViolet Sensor
UV	Ultraviolet
VIPs	Vacuum Insulated Panels
WBS	Work Breakdown Structure
WT	Weight Percentage
XPS	X-Ray Photoelectron Spectroscopy
XPS	Extruded Polystyrene
ZARM	Zentrum für Angewandte Raumfahrttechnologie und Mikrogravitation

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APPENDIX A - EXPERIMENT REVIEWS

Preliminary Design Review – PDR

	
REXUS / BEXUS Experiment Preliminary Design Review	
Flight: BEXUS 38/39	
Payload Manager: TBC	
Experiment: COATHEAL	
Location: ESRANGE	Date: 29 th January 2026
Experiment Team members - Mirică Eduard Leonard - Cîrja Rareș - Drăgoi Cristian-Theodor - Tănase Andreea Luiza	Panel Jorge Galván Lobo, Esmée Menting, Ganesh Voggu, Maximilian Nuernberger, Koen Debeule, Alessandro Billi, Michael Becker, Simon Mawn, Merle Cornelius, Ava Pohley, Leonard Kobow, Moritz Aicher, Sebastian Dominik, Sebastian Weiss, Maximilien Hiepp
Panel Comments and Recommendations ▪ Editorial - ACTION: correct date in change record - ACTION: Experiment platform is the stratospheric balloon, when referring to your experiment you may use experiment or experiment assembly. - ACTION: rewrite the document especially the sections which are heavily created by AI. Only use AI for spell check or sentence fluidity. Do not use AI for generative purpose. The SED should be short and to the point, AI generated text usually does the opposite. - ACTION: Put the reference number next to the sentence where you are using it, not condensed at the end of the paragraph. - ACTION: abstract, scientific background and experiment concept are different parts of the document. Fill them accordingly (e.g., a deeper analysis of literature and state of the art on scientific background + the content of chapter 4.10). ▪ Science and objectives - ACTION: modify your objectives, centring your aim to observe and demonstrate the performances of self-healing mechanism. How belongs to functional requirements. - ACTION: further elaborate the mean of generation of the crack, confirm the effectiveness of the thermal stress, justify the exclusion of the mechanical component. Research a suiting sample that has highest susceptibility to the mean you want to apply and higher tolerance to other stresses. - ACTION: consider if a mechanism is needed to create cracks in-situ. - ACTION: develop an in-flight observation method, whether it is an optical visualization or laser/electric flow/ other. Expand the research on possible alternatives, do not limit your evaluation to saying that a microscope will not fit. It can also be focused on only one reference sample. ▪ Requirements and constraints (SED chapter 2) - ACTION: Rewrite the section based on what you learned during the training week and the following notes. - ACTION: F7-8 not functional. - ACTION: add requirements regarding the heater. - ACTION: add requirements for the microcrack generation. - ACTION: add all design requirements, shock of vehicle, data and power supply, voltage, current, data rate. - ACTION: C1 is not a constraint - ACTION: C2 remove	

- **Remark:** O2 requirement is binding on the procedure, evaluate if that is needed or if it can be differently.
- **ACTION:** split F3-4-5, to measure and to record.
- **ACTION:** F6 add "each" after "from"
- **ACTION:** P1 to P3 split accuracy and range requirements
- **ACTION:** add performance requirements between
- **ACTION:** add weight and power requirements
- **ACTION:** estimate the data and replace the TBDs
- **Mechanics** (SED chapter 4.2.1 & 4.4)
 - **ACTION:** Update Figure 4.1.1 to show the connection between COATHEAL and BPS
 - **ACTION:** 4.2 adjust the mounting to be connected to both sides of ESCARGO
 - **ACTION:** re evaluate the choice of honeycomb structure against different design choices (e.g., cutout with more bolts) for manufacturing time and easiness.
 - **ACTION:** elaborate the study on the sample preservation procedure for landing.
 - **Remark:** evaluate the possibility to use COTS when possible, trade-off with design needs.
 - **Remark:** assess if the bolts inside the box are reachable.
- **Electronics and data management** (SED chapter 4.2.2, 4.2.3, 4.5, 4.7, 4.9)
 - **ACTION:** Investigate the data generation and loss by the photodiode, consider including an amplifier.
 - **ACTION:** rewrite the power budget, list components not only subsystems.
 - **ACTION:** rewrite power consumption over time, accounting for 10 hours.
 - **Remark:** research lower quality and less expensive DCDC
 - **ACTION:** include description of the battery mentioned in other sections and its specs.
- **Thermal** (SED chapter 4.2.4 & 4.6)
 - **Remark:** convection at high altitude is negligible. Good to use it for worst case if it is negative, but it is not necessary.
 - **ACTION:** evaluate insulation to gain margin in the power budget (aid to the heaters).
 - **ACTION:** add requirement for broad daylight if needed.
 - **Remark:** you can investigate the black frame possibility for natural heating.
 - **ACTION:** define your thermal design, including the balance between passive and active instruments.
- **Software** (SED chapter 4.8)
 - **ACTION:** Sync the GNSS with the internal clock with GPSD.
 - **ACTION:** clear flowchart including automatic and manual telecommands.
- **Verification and testing** (SED chapter 5)
 - **ACTION:** mechanical test: gently load is wrong. Mechanical test shall be of 10 g and 5 g, with proper procedures and report.
 - **ACTION:** adjust inconsistencies between the requirements and the verifications.
 - **ACTION:** F1 remove I
 - **ACTION:** F3 to F6 use R & T, remove A when possible.
 - **ACTION:** add estimate date for the verification.
 - **ACTION:** performance requirement are not to be verified by T or A, use R when possible.
- **Safety and risk analysis** (SED chapter 3.5)
 - **ACTION:** adjust severity too high in some risks
 - **ACTION:** Add re-evaluation of risk after mitigation by adding a new risk with new SxP and new action (PE10 evolves in PE11, keep them both)
 - **Remark:** PE10 severity 3 forces you to have redundant team members, but you state that you are comfortable with the number of members.
- **Launch and operations** (SED chapter 6)
 - **ACTION:** clearly state if you need to do chemical preparation on your samples during Campaign.
 - **ACTION:** elaborate a more in-depth and clearer version of the requirements for post-flight recovery.
 - **ACTION:** mark the requirement for the sun: daylight is a hard requirement; position of the sun is a nice to have.

- **ACTION:** adjust flight profile to fit 10 hours.
- Organisation, project planning & outreach (SED chapters 3.1, 3.2, 3.3 & 3.4)
 - **ACTION:** rewrite the section without the contribution of AI.
 - **ACTION:** add the visual representation of WBS (can be in appendix, must be clear comprehensive and readable, HD)
 - **ACTION:** add interdependencies between tasks and milestones in your schedule.
 - **ACTION:** investigate the balance between workload and workforce in May and June, adding contingency plan if necessary.
 - **ACTION:** design the logo with higher quality.
 - **ACTION:** the timeline for the creation of a society does not fit the project timeline, address procurement specifics: costs and time. Program when you will need to buy, make sure that the funding is available.

Internal Panel Discussion

- Summary of main actions for the experiment team
 - The main concern of the panel is about the science. You have accepted to not have any preliminary results by your experiment and relying solely on observation performed after the flight, the landing, the storage, and the transportation of the samples.
 - **ACTION:** further elaborate the mean of generation of the crack, confirm the effectiveness of the thermal stress, justify the exclusion of the mechanical component. Research a suiting sample that has highest susceptibility to the mean you want to apply and higher tolerance to other stresses.
 - **ACTION:** develop an in-flight observation method, whether it is an optical visualization or laser/electric flow/other. Expand the research on possible alternatives, do not limit your evaluation to saying that a microscope will not fit. It can also be focused on only one reference sample.
 - **ACTION:** consider if a mechanism is needed to create cracks in-situ.
 - A second point of concern is the quality of the SED, with some parts poorly compiled and others AI-generated.
 - **ACTION:** abstract, scientific background and experiment concept are different parts of the document. Fill them accordingly (e.g., a deeper analysis of literature and state of the art on scientific background + the content of chapter 4.10).
 - **ACTION:** Review and rewrite the document in higher quality with further details.
- Accommodation: BEXUS 38/39
- PDR Result: **Conditional Pass**
- Next SED version SED V1-1 due: 08th March 2026
- Next Action Matrix version V1-1 due: 29th March 2026

Critical Design Review – CDR

Integration Progress Review – IPR

Experiment Acceptance Review – EAR

Appendix B – OUTREACH AND MEDIA COVERAGE

APPENDIX C – ADDITIONAL TECHNICAL INFORMATION

APPENDIX D – CHECKLISTS